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Chapter 1

Recognition of Need

1.1 Introduction

A system is a structured collection of interdependent components that operate together to accomplish a defined objective. In the field of System Analysis and Design, the recognition of need is considered the initial and most essential stage, because it establishes the justification for examining an existing system and identifying the areas that require improvement. At this stage, the analyst studies the current environment, observes operational shortcomings, and determines the gaps between present performance and the expected standard of service. The purpose of this process is not merely to detect errors, but to understand the underlying causes of inefficiency and to provide a foundation for systematic improvement in later stages of analysis and design.

In modern society, digital platforms have become deeply integrated into everyday activities, particularly in transportation, food delivery, and logistics services. Among the most widely used digital service platforms in Bangladesh, Pathao occupies a significant position due to its convenience, accessibility, and broad range of services. The application supports ride-sharing, parcel delivery, food delivery, and related on-demand services through a single mobile platform. Because of this multi-service structure, Pathao serves a large and diverse user base, including passengers, delivery users, drivers, and service providers. Its success depends not only on operational scale, but also on the reliability, efficiency, and usability of the system.

Although Pathao has established itself as one of the leading digital service platforms in the country, the system is not free from limitations. As with many large-scale service applications, users may encounter technical, functional, and service-related difficulties while interacting with the platform. These issues may arise in areas such as application responsiveness, location tracking, booking stability, fare transparency, and customer support. When such problems occur, they can reduce user satisfaction, create confusion, and weaken trust in the service. Since Pathao is used in situations that often demand quick responses and dependable outcomes, even minor system weaknesses may have a noticeable impact on overall service quality.

From a system analysis perspective, Pathao presents an important and relevant subject for study because it reflects the challenges faced by a modern, real-time digital service platform operating in a competitive environment. The need for improvement may arise not only from user complaints, but also from the growing expectations of customers who demand speed, accuracy, and consistency from digital applications. Therefore, it is necessary to analyze Pathao in a structured way in order to identify the existing needs of the system and determine what kinds of improvements would enhance its overall effectiveness.

This report is prepared as a single continuous academic project that will be developed gradually over several weeks. Each chapter will build upon the previous one so that the final report remains logically connected and methodically organized. The first chapter introduces the recognition of need, identifies the main weaknesses of the current system, and establishes the basis for the later chapters of feasibility study, analysis, and design. In this way, the report will maintain both academic coherence and practical relevance throughout the entire project.

1.2 Problem Identification

The identification of problems is a fundamental part of system analysis because it reveals the operational weaknesses that prevent a system from functioning at its full potential. A system may appear successful from the outside, yet still contain internal issues that reduce efficiency, user satisfaction, and service dependability. In the case of Pathao, several important problem areas can be recognized through observation of the user experience and the nature of the services provided. These issues are described below.

Problem 01: Inconsistent Application Performance

One of the most visible issues in a digital service platform is inconsistent application performance. For Pathao, users may sometimes face slow loading pages, temporary lag, or delays while navigating between different sections of the app. Such problems may appear small in isolation, but in a service that is expected to operate quickly and smoothly, these delays can create significant inconvenience. Users often rely on Pathao when they need immediate transportation or fast delivery services, and any interruption in application speed may reduce the efficiency of the entire service process.

In addition, unstable application performance can negatively influence user confidence. A user who repeatedly experiences delay or unresponsiveness may begin to question the reliability of the platform. This is particularly important in a mobile service environment, where customers expect near-instant access and minimal friction. Therefore, performance consistency is not

merely a technical requirement, but a critical factor in maintaining service quality and customer retention.

Problem 02: Location Tracking and Map Accuracy Issues

Location-based accuracy is one of the core functions of a ride-sharing application. Pathao depends heavily on GPS tracking and digital maps for ride booking, route display, and driver-customer coordination. If the map shows an incorrect route, the pickup point is displayed inaccurately, or the live location fails to update properly, the result may be confusion for both the customer and the driver. These errors can cause late arrivals, unnecessary waiting time, and avoidable frustration.

In a platform where movement and timing are central to the service model, tracking errors can have serious operational implications. The customer may be unable to locate the driver properly, while the driver may reach the wrong place or waste time searching for the correct pickup point. In such cases, the quality of the overall service is reduced even if the ride itself is completed successfully. As a result, accurate location tracking must be treated as a major system requirement and a key area of improvement.

Problem 03: Booking Reliability and Cancellation Problems

A service platform must ensure that users can complete a booking with confidence. In Pathao, one possible concern is the reliability of the booking and cancellation process. If a booking is accepted and later canceled unexpectedly, or if the assigned driver takes too long to arrive, the user experience becomes unstable and uncertain. Repeated disruptions of this kind can lead users to lose trust in the platform's ability to deliver consistent service.

Booking reliability is particularly important because users often choose Pathao when they are in a hurry or need transportation at a specific time. In such situations, delays or cancellations do not merely cause inconvenience; they can directly affect the user's schedule, commitments, and overall satisfaction. Therefore, any weakness in the booking workflow should be considered a significant issue in the system analysis process.

Problem 04: Fare Transparency and Pricing Confusion

Fare transparency is another important concern in digital transportation services. Users expect to know how fares are calculated, what additional charges may apply, and what amount will ultimately be paid. If the fare displayed at the beginning of the trip differs from the final amount without a clear explanation, users may feel confused or dissatisfied. Similarly, if surge pricing,

service fees, or payment totals are not explained clearly, the system may appear unclear or unfair to the customer.

In service platforms such as Pathao, trust is strongly influenced by financial transparency. Customers are more likely to continue using a platform when they feel that pricing is predictable and understandable. Conversely, uncertainty in fare calculation may create dissatisfaction and reduce the perceived fairness of the service. For this reason, pricing clarity is a key issue that deserves careful attention in system analysis.

Problem 05: Customer Support and Complaint Resolution Delays

No service platform can operate effectively without a dependable customer support mechanism. In the case of Pathao, users may need assistance for ride-related issues, payment problems, delivery concerns, or account-related difficulties. If the support system is slow, difficult to navigate, or unable to provide timely solutions, users may feel abandoned by the service. The speed and quality of complaint resolution are therefore essential to maintaining user trust.

When customer support is delayed, the impact extends beyond the individual complaint. A dissatisfied customer may share negative feedback with others, reducing confidence in the platform more broadly. Furthermore, unresolved issues may accumulate and create operational pressure on the support system. Therefore, complaint handling and user assistance must be considered an integral part of the overall system effectiveness.

1.3 Conclusion

The recognition of the need phase demonstrates that Pathao, despite being a highly popular and widely used digital service platform in Bangladesh, still contains several areas that may require systematic improvement. The observed issues include inconsistent application performance, location tracking inaccuracies, booking reliability problems, fare transparency concerns, and delays in customer support. Each of these issues can affect the user experience in a meaningful way and may reduce confidence in the platform if not addressed properly.

From a system analysis and design standpoint, these problems justify the need for a detailed study of Pathao's current operations. The objective of such a study is not to criticize the platform, but to identify the practical limitations that exist within the present system and to determine how they may be improved through structured analysis and design solutions. By

recognizing these needs at the beginning of the project, the report establishes a strong academic and analytical foundation for the chapters that will follow.

This chapter therefore serves as the starting point of the entire project. It defines the scope of the study, identifies the key problem areas, and creates a logical basis for further evaluation in the upcoming phases of feasibility analysis, system analysis, and design development. A well-structured response to these needs will contribute to a more efficient, reliable, and user-centered Pathao system, which is consistent with the goals of modern system analysis and design.

Chapter 2

Initial Feasibility Study

2.1 Introduction

Following the recognition of need presented in Chapter 1, the next phase of the System Development Life Cycle (SDLC) is the Initial Feasibility Study. This phase is conducted to evaluate whether the proposed system improvement project is practical, beneficial, and achievable before investing significant resources into detailed system analysis and design. An initial feasibility study provides decision-makers and system analysts with an early understanding of the project's viability by examining the current operational environment, identifying existing limitations, and assessing the possibility of implementing effective solutions.

The primary objective of an initial feasibility study is to determine whether the identified problems can be addressed using available technological resources, organizational capabilities, financial support, and operational procedures. Rather than proposing a complete solution, this phase focuses on evaluating whether the project should proceed to the next stages of development. A successful feasibility study minimizes project risks, reduces unnecessary expenditures, and ensures that future development efforts are aligned with organizational objectives and user requirements.

In the context of this project, the selected organization is **Pathao Limited**, one of the leading digital technology companies in Bangladesh. Pathao has significantly transformed the country's transportation and logistics industry by introducing technology-driven ride-sharing, food delivery, parcel delivery, and courier services through a single mobile application. Since its establishment, the company has experienced rapid growth and has become an essential digital platform serving millions of users across the country.

Despite its success, continuous expansion and increasing customer expectations have introduced several operational and technical challenges. As discussed in Chapter 1, five major issues have been identified within the existing Pathao system: inconsistent application performance, location tracking and map accuracy issues, booking reliability and cancellation problems, fare transparency and pricing confusion, and delays in customer support and

complaint resolution. These issues have varying degrees of impact on user satisfaction, operational efficiency, and overall service reliability.

Therefore, before proposing any modifications to the existing system, it is necessary to conduct an initial feasibility study to determine whether improvements are realistic from technical, operational, economic, and managerial perspectives. The findings of this chapter will establish the foundation for the detailed system analysis presented in the following chapter and will help justify the continuation of the proposed system improvement project.

2.2 Existing System of Pathao

Pathao Limited operates a centralized, technology-driven service platform that integrates multiple on-demand services into a single mobile application. The current system is designed to connect customers, service providers, merchants, delivery personnel, and administrators through cloud-based infrastructure supported by mobile computing, Global Positioning System (GPS) technology, internet connectivity, and digital payment services. This integrated architecture enables users to access different services from one application while allowing the organization to manage large volumes of transactions efficiently.

The operation of the existing system begins with user registration. New users create an account using their mobile phone number and complete the authentication process through One-Time Password (OTP) verification. Once registration is completed, users gain access to all available Pathao services without requiring separate accounts for each service category. This integrated account management approach improves user convenience while simplifying customer data management for the organization.

For ride-sharing services, users specify their pickup location and destination using GPS-enabled digital maps. Based on these inputs, the system estimates the travel fare, identifies nearby drivers, and forwards the ride request to the most appropriate service provider. After a driver accepts the request, both the customer and the driver can monitor each other's live location until the ride is completed. At the conclusion of the journey, payment is processed through either cash or available digital payment methods, followed by a mutual rating and feedback process.

The food delivery module follows a similar operational workflow. Customers browse participating restaurants, review available menus, place orders through the application, complete payment, and monitor the delivery process through real-time tracking. Likewise, the parcel and courier modules allow users to schedule package collection, specify delivery

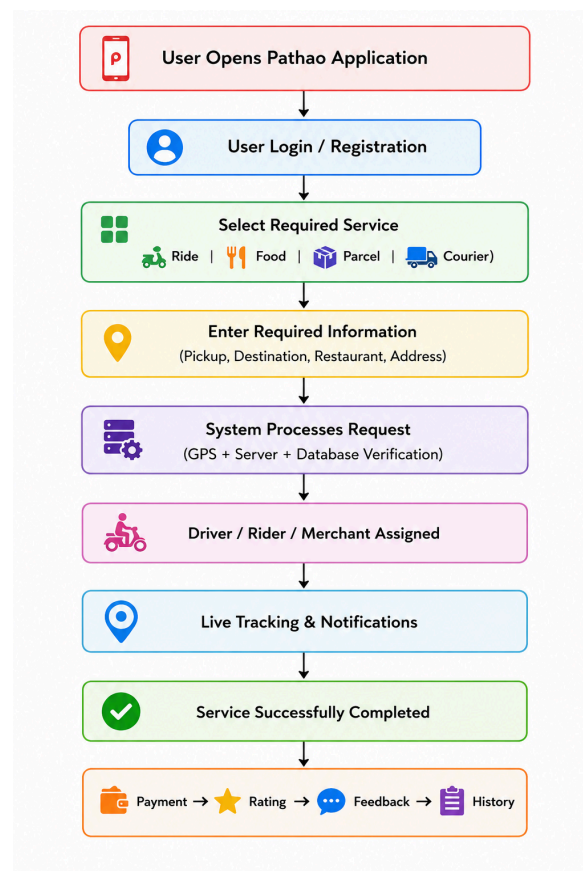
Pathao App System Analysis & Design

destinations, and track shipment progress until successful delivery. These integrated services have enabled Pathao to establish itself as one of Bangladesh's leading digital lifestyle platforms.

The existing system also incorporates several supporting functionalities, including transaction history, promotional campaigns, customer notifications, account management, customer support, and service rating mechanisms. These features contribute to a comprehensive digital ecosystem that enhances customer convenience and supports the organization's operational efficiency.

Although the current system successfully performs its primary functions, the increasing number of users and the complexity of daily operations have exposed several limitations that affect overall service quality. Performance inconsistency, occasional GPS inaccuracies, ride cancellation issues, pricing confusion, and delays in customer support continue to influence customer satisfaction and operational effectiveness. These challenges indicate that while the current system is functionally successful, there remains considerable scope for improvement through systematic analysis and redesign.

Figure 2.1: General Workflow of the Existing Pathao System



The existing workflow demonstrates that the Pathao platform integrates multiple interconnected subsystems within a unified digital environment. While this architecture provides convenience and scalability, it also increases the complexity of system management. Any weakness within one subsystem, such as inaccurate location tracking or application performance issues may influence the effectiveness of the entire platform. Consequently, a systematic feasibility evaluation of the identified problems is necessary before proposing any modifications to the existing system.

2.3 Initial Feasibility Study

The purpose of the initial feasibility study is to evaluate whether the identified problems within the existing Pathao system can be addressed through practical and achievable system improvements. At this stage, the objective is not to develop the complete solution but to determine whether each problem is significant enough to justify further analysis and whether the organization possesses the necessary technical capability, operational readiness, financial resources, and managerial support to implement improvements successfully.

Based on the findings presented in Chapter 1, five major problems have been identified within the existing Pathao system. Each problem has been analyzed individually to understand its current impact on users and organizational performance while determining the feasibility of future system enhancement.

2.3.1 Problem 01: Inconsistent Application Performance

Existing Situation

The performance of a mobile application is one of the most important indicators of service quality in any digital platform. Since Pathao provides real-time transportation, food delivery, and parcel services, users expect the application to respond immediately while navigating through different features. However, during periods of high demand or unstable network conditions, users may occasionally experience slow loading, delayed responses, or temporary freezing while using different modules of the application.

These performance inconsistencies may occur during ride booking, restaurant browsing, payment confirmation, or while switching between different services. Although these issues do not always prevent users from completing their activities, they reduce the overall responsiveness of the application and negatively influence the user experience.

As the number of registered users continues to increase, the existing infrastructure is required to process a significantly larger number of simultaneous requests. Without continuous optimization of server resources, application architecture, and database communication, maintaining consistent performance becomes increasingly challenging.

Impact on the Existing System

Application performance directly influences customer satisfaction because users generally access Pathao when immediate service is required. Delays of even a few seconds during ride booking or payment confirmation may create frustration and reduce user confidence in the platform.

From an organizational perspective, unstable application performance may increase customer complaints, reduce successful transaction rates, and create additional workload for customer support representatives. Poor performance also affects the organization's reputation, as users often compare the responsiveness of different digital service platforms before selecting their preferred application.

Initial Technical Feasibility

The improvement of application performance is considered technically feasible because the existing Pathao platform already operates using modern cloud infrastructure, mobile technologies, and scalable software architecture. Performance optimization can be achieved through database optimization, improved server load balancing, efficient API communication, application code optimization, and better resource management without requiring complete redevelopment of the existing system.

Since these improvements primarily involve software enhancement rather than hardware replacement, implementation can be integrated into the current development cycle with manageable technical complexity.

Initial Operational Feasibility

Operationally, the proposed improvements are highly feasible because users will continue interacting with the same application interface while experiencing improved speed and responsiveness. No significant changes in user behavior or employee responsibilities will be required. Therefore, organizational resistance to implementation is expected to be minimal.

Initial Economic Feasibility

Although performance optimization requires investment in software engineering, infrastructure monitoring, and testing, the expected benefits significantly outweigh the implementation costs. Improved performance can reduce customer complaints, increase successful transactions, improve customer retention, and strengthen the organization's competitive position. Consequently, the proposed improvement is considered economically feasible.

Preliminary Recommendation

The existing application should undergo continuous performance monitoring and optimization through infrastructure enhancement, server resource balancing, efficient database management, and periodic software updates. These improvements are expected to provide a more stable and responsive user experience while supporting future growth of the Pathao platform.

2.3.2 Problem 02: Location Tracking and Map Accuracy Issues

Existing Situation

Location tracking represents one of the most fundamental components of the Pathao service platform. Every ride request, parcel delivery, and food delivery operation depends on accurate GPS positioning and real-time communication between customers and service providers. The existing system utilizes digital maps and Global Positioning System (GPS) technology to determine pickup locations, calculate estimated routes, assign nearby drivers, and monitor ongoing trips.

Despite the effectiveness of the overall navigation system, occasional inaccuracies in GPS positioning or delayed map updates may occur due to network limitations, environmental conditions, or device-specific factors. These situations may cause discrepancies between the displayed location and the user's actual position, resulting in confusion during pickup and delivery operations.

Such inaccuracies become particularly problematic in densely populated urban areas where multiple roads, buildings, and traffic conditions affect GPS precision. As Pathao continues expanding its services across Bangladesh, maintaining consistent location accuracy becomes increasingly important for ensuring reliable service delivery.

Impact on the Existing System

Location inaccuracies directly affect operational efficiency because both customers and drivers depend on real-time navigation throughout the service process. Incorrect pickup locations may increase waiting time, delay service completion, and reduce customer satisfaction. Drivers may also experience unnecessary travel distance, increased fuel consumption, and reduced operational efficiency when location data is inaccurate.

From the organizational perspective, repeated navigation issues may increase complaint frequency, reduce customer trust, and influence users to consider alternative service platforms that provide more accurate navigation experiences.

Initial Technical Feasibility

Improving location accuracy is technically feasible because the required enhancements primarily involve optimization of existing GPS algorithms, map synchronization, route calculation methods, and location update frequency. Since Pathao already utilizes advanced navigation technologies, improvements can be implemented through software upgrades and integration with more accurate mapping services rather than complete system replacement.

Initial Operational Feasibility

The proposed improvements can be introduced without changing the overall workflow of customers or drivers. Users will continue using the existing application interface while benefiting from more accurate navigation and improved pickup coordination. Therefore, operational feasibility is considered high.

Initial Economic Feasibility

Although improvements may require investment in mapping services, software optimization, and system testing, the long-term operational benefits, including reduced delays, improved customer satisfaction, and increased service efficiency justify the required expenditure. Consequently, this improvement is considered economically feasible.

Preliminary Recommendation

Pathao should continuously improve its location-based services by optimizing GPS processing, increasing map update accuracy, refining route calculation algorithms, and monitoring navigation performance through regular system evaluation. These improvements will significantly enhance overall service reliability and customer confidence.

2.3.3 Problem 03: Booking Reliability and Cancellation Problems

Existing Situation

Booking reliability is one of the most critical performance indicators of an online ride-sharing platform because it directly influences customer confidence and service dependability. The existing Pathao system is designed to automatically identify nearby drivers and assign ride requests based on factors such as location, availability, and estimated response time. Under normal operating conditions, this process enables users to obtain transportation quickly and conveniently. However, there are situations in which the booking process becomes less reliable due to delayed driver acceptance, ride cancellations after confirmation, or prolonged waiting times before a driver reaches the pickup location.

These situations may arise because of heavy traffic conditions, high service demand during peak hours, limited driver availability, or operational decisions made by individual drivers. Although the existing system attempts to reassign rides after cancellations, repeated interruptions increase uncertainty and reduce the effectiveness of the booking mechanism. Since customers generally use Pathao when timely transportation is required, consistency throughout the booking process is essential for maintaining service quality.

Impact on the Existing System

Unreliable booking and frequent cancellations affect both customers and the organization. Customers may experience delays in reaching their destinations, disruption of personal or professional schedules, and reduced confidence in the platform. In emergency or time-sensitive situations, repeated booking failures can significantly decrease customer satisfaction.

From an organizational perspective, booking instability increases customer complaints, places additional pressure on customer support, and may reduce customer retention. If users perceive that rides cannot be obtained reliably, they may choose competing ride-sharing platforms, ultimately affecting the organization's market competitiveness and long-term business performance.

Initial Technical Feasibility

Improving booking reliability is technically feasible because the required enhancements primarily involve optimization of the existing ride allocation algorithms, driver assignment logic, cancellation management procedures, and demand forecasting techniques. The current infrastructure already supports automated ride matching; therefore, these improvements can be implemented through software modifications without requiring major architectural changes.

Initial Operational Feasibility

The proposed improvements can be integrated into the existing operational workflow without introducing significant changes for customers or drivers. Users will continue requesting rides through the current application interface, while the improved system will provide more reliable ride allocation and reduce unnecessary cancellations. Consequently, operational feasibility is considered high.

Initial Economic Feasibility

Although algorithm optimization, testing, and system monitoring require additional development resources, the resulting improvement in customer satisfaction and ride completion rates is expected to increase revenue while reducing operational losses associated with cancellations and complaint handling. Therefore, the proposed enhancement is economically feasible.

Preliminary Recommendation

The booking management system should be strengthened through improved driver allocation algorithms, intelligent demand prediction, automated reassignment mechanisms, and continuous monitoring of cancellation patterns. These improvements will increase booking reliability and enhance customer confidence in the Pathao platform.

2.3.4 Problem 04: Fare Transparency and Pricing Confusion

Existing Situation

Transparent pricing plays an important role in establishing customer trust within digital transportation services. The existing Pathao system estimates fares using multiple factors, including travel distance, estimated travel time, traffic conditions, service type, and demand fluctuations. While this pricing mechanism enables dynamic fare calculation, users may occasionally experience uncertainty regarding the final payable amount, particularly during periods of high demand or when additional service charges are applied.

Customers generally expect that the estimated fare displayed before confirming a booking will closely represent the amount paid after the service is completed. Any noticeable variation between estimated and final charges, if not explained clearly, may create confusion and reduce confidence in the fairness of the pricing system. Furthermore, promotional offers, discount campaigns, and digital payment incentives sometimes introduce additional complexity into the

pricing process. Without sufficiently clear explanations, users may misunderstand how the final fare has been determined.

Impact on the Existing System

Pricing uncertainty can influence customer satisfaction even when the transportation service itself is completed successfully. If users perceive the pricing process to be unclear or inconsistent, they may hesitate to continue using the platform. Financial transparency is particularly important because payment directly influences customer trust.

From the organizational perspective, unclear pricing may increase customer complaints, refund requests, and inquiries directed toward customer support. These additional operational activities consume organizational resources and may negatively affect the company's public reputation.

Initial Technical Feasibility

The improvement of fare transparency is technically feasible because it mainly requires modifications to the existing fare calculation interface rather than redevelopment of the pricing engine. Additional fare breakdowns, improved user notifications, and clearer explanations of pricing components can be incorporated into the current application through software updates.

Initial Operational Feasibility

Operational implementation is straightforward because users will continue using the same booking procedure while receiving more detailed pricing information before confirming their requests. No major procedural changes will be required for drivers, administrators, or customers.

Initial Economic Feasibility

The financial investment required for improving pricing transparency is relatively small compared with the potential benefits. Increased customer trust, reduced pricing disputes, lower customer support workload, and improved customer retention collectively justify the implementation cost. Therefore, the proposed improvement is economically feasible.

Preliminary Recommendation

The Pathao application should present a more comprehensive fare breakdown before ride confirmation by clearly displaying estimated fare components, service charges, promotional discounts, and any applicable dynamic pricing adjustments. Greater pricing transparency will improve customer understanding and strengthen confidence in the platform.

2.3.5 Problem 05: Customer Support and Complaint Resolution Delays

Existing Situation

Customer support serves as an essential component of any service-oriented digital platform because it provides assistance when users experience operational difficulties. The existing Pathao system offers customer support through in-application reporting mechanisms and other communication channels for resolving issues related to rides, food delivery, parcel services, payments, refunds, and account management.

Although the existing support infrastructure successfully resolves many customer concerns, the increasing number of users has created additional pressure on customer service operations. During periods of high service demand, some complaints may require longer processing times before appropriate solutions are provided. Delays in complaint resolution can reduce customer confidence, particularly when issues involve financial transactions or unsuccessful service experiences.

Efficient complaint management has become increasingly important because customer expectations continue to rise as digital service platforms expand throughout Bangladesh.

Impact on the Existing System

Slow customer support directly affects user satisfaction because unresolved problems often remain associated with negative service experiences. Customers who cannot obtain timely assistance may become reluctant to use the platform again, even if the original issue was relatively minor.

From an organizational perspective, delayed complaint resolution may increase negative customer reviews, reduce brand reputation, and create additional operational workload due to repeated follow-up inquiries. Maintaining efficient customer support is therefore essential for protecting both customer loyalty and organizational credibility.

Initial Technical Feasibility

The proposed improvements are technically feasible because the existing customer support infrastructure can be enhanced through workflow automation, intelligent ticket management, chatbot integration, and improved internal support systems. These enhancements can be implemented without replacing the existing customer service platform.

Initial Operational Feasibility

Operational feasibility is considered high because the proposed improvements strengthen existing business processes rather than replacing them. Customer service representatives will continue performing their current responsibilities while benefiting from improved support tools and faster issue management.

Initial Economic Feasibility

Although implementation requires investment in software enhancement, customer service technologies, and employee training, the expected reduction in complaint handling time, improvement in customer retention, and enhancement of organizational reputation provide substantial long-term benefits. Therefore, the proposed improvements are considered economically feasible.

Preliminary Recommendation

Pathao should strengthen its customer support operations by implementing intelligent ticket management systems, improving response monitoring, integrating automated support technologies, and establishing performance indicators for complaint resolution. These initiatives will improve service quality while increasing customer satisfaction and long-term user loyalty.

2.4 Summary of the Initial Feasibility Study

The initial feasibility assessment demonstrates that each of the five identified problems can be addressed through practical and achievable system improvements. The existing technological infrastructure, operational framework, and organizational capabilities of Pathao provide a strong foundation for implementing these enhancements without requiring complete redevelopment of the existing platform. Although each improvement requires investment in software optimization, testing, monitoring, and employee support, the anticipated benefits, including improved customer satisfaction, increased operational efficiency, reduced complaint

rates, and stronger market competitiveness significantly outweigh the associated implementation costs.

Based on the findings of this chapter, the proposed system improvement project is considered technically feasible, operationally acceptable, economically justifiable, and suitable for further investigation during the detailed system analysis phase. The following table summarizes the overall feasibility evaluation.

Table 2.1: Summary of Initial Feasibility Assessment

Identified Problem	Technical Feasibility	Operational Feasibility	Economic Feasibility	Overall Assessment
Inconsistent Application Performance	High	High	High	Feasible
Location Tracking and Map Accuracy Issues	High	High	High	Feasible
Booking Reliability and Cancellation Problems	High	High	High	Feasible
Fare Transparency and Pricing Confusion	High	High	High	Feasible
Customer Support and Complaint Resolution Delays	High	High	High	Feasible

2.5 Conclusion

The Initial Feasibility Study confirms that the proposed improvement project for the Pathao platform is both practical and beneficial. The analysis indicates that the identified operational problems do not arise from fundamental weaknesses in the overall system architecture but rather from specific functional areas that require continuous optimization and refinement. Since the existing platform already possesses a mature technological infrastructure and an established operational framework, implementing targeted improvements is considered both realistic and achievable.

The feasibility assessment further demonstrates that the proposed enhancements can improve system performance, operational efficiency, and customer satisfaction while maintaining compatibility with the organization's existing business processes. From technical, operational, and economic perspectives, the project satisfies the essential conditions required to proceed to the next phase of the System Development Life Cycle.

The outcomes of this chapter provide the necessary justification for conducting a comprehensive system analysis in the following chapter. Chapter 3 will focus on detailed information gathering, analytical techniques, process modeling, and root cause analysis to identify appropriate solutions for the problems recognized during the feasibility study.

Chapter 3

Analysis

3.1 Introduction

System analysis is a fundamental stage of the System Development Life Cycle (SDLC) in which an existing system is examined systematically to identify its processes, information requirements, operational limitations, user expectations, and opportunities for improvement. The purpose of this phase is not merely to identify what is wrong with the existing system, but to understand why the problems occur, how they affect different stakeholders, and what type of changes may be required to improve the overall system.

In the previous chapters, the need for analyzing the Pathao application was established and an initial feasibility study was conducted. The study identified five major problems associated with the existing system: inconsistent application performance, location tracking and map accuracy issues, booking reliability and cancellation problems, fare transparency and pricing confusion, and customer support and complaint resolution delays. These problems are not isolated from one another. Rather, they are interconnected through the application's service workflow. For example, an application performance problem can interfere with map loading or booking requests, while inaccurate location information can affect driver matching and eventually contribute to cancellations. Similarly, an unsuccessful booking or pricing dispute may generate additional pressure on the customer-support system.

The analysis phase therefore focuses on examining the existing Pathao system from both a functional and user-oriented perspective. The investigation considers the major stakeholders involved in the service process, including customers, drivers or delivery partners, merchants where applicable, system administrators, payment services, and customer-support personnel. The analysis also considers the interaction between the mobile application, location services, application servers, databases, notification mechanisms, and transaction-processing components.

To obtain a broader understanding of the existing system, several information-gathering techniques are considered. These include review of relevant literature and application procedures, application-based observation, structured stakeholder interviews, and user

questionnaires. The information obtained through these techniques is then represented using tables, charts, process models, and Data Flow Diagrams (DFDs). The reference report similarly treats information gathering as a systematic stage involving literature review, procedures, forms, observation, interviews, and questionnaires.

Following information gathering, a cost-benefit analysis is performed to assess whether the proposed improvements are economically reasonable. Finally, detailed feasibility analysis is conducted through root-cause analysis and the development of alternative candidate systems. The candidate systems are compared using qualitative and weighted evaluation criteria, allowing the most appropriate improvement strategy to be selected for the subsequent design phase.

3.2 Information Gathering

Information gathering is one of the most important activities in system analysis because an effective solution cannot be designed without first understanding the existing system accurately. The analyst must collect information regarding the way users interact with the system, the sequence of operations, the data generated during each transaction, the difficulties encountered by stakeholders, and the limitations of the existing processes.

The information-gathering process for this project is designed specifically around the Pathao application. Since Pathao is a large commercial organization and its internal databases, source code, server architecture, financial records, and management-level operational information are not publicly available for this academic project, the analysis does not claim access to confidential organizational information. Instead, it focuses on publicly observable application functions, documented service procedures, structured observation, user-oriented investigation, and analytical modeling.

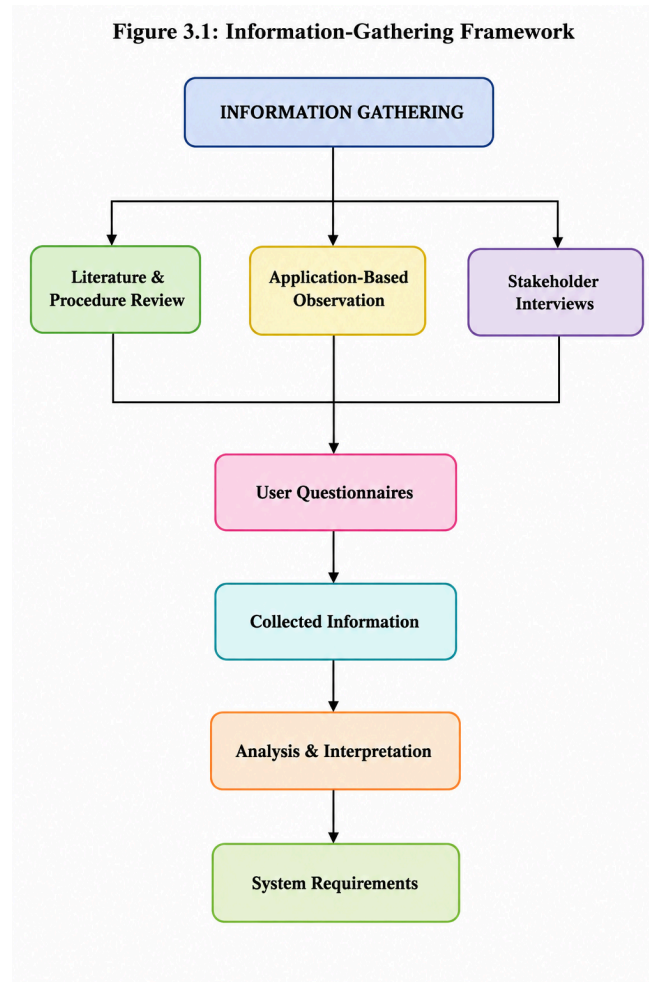
The major information-gathering methods used for this project are:

1. Review of Literature, Procedures and Application Functions
2. Application-Based Observation
3. Stakeholder Interviews
4. User Questionnaires

These methods complement one another. Literature and procedure review provides background information, observation examines the practical workflow, interviews provide qualitative

stakeholder perspectives, and questionnaires provide structured information that can be converted into quantitative analysis.

Figure 3.1: Information-Gathering Framework



3.2.1 Review of Literature, Procedures and Application Functions

Reviewing relevant literature and system procedures provides the initial knowledge required to understand the existing Pathao environment. In system analysis, this activity helps the analyst identify the major functions of the system, understand how different processes are connected, and determine which areas require further investigation.

The review for this project concentrates on the service processes that are directly related to the five problems identified in Chapter 1.

3.2.1.1 Review of Application Functions

The Pathao application provides a centralized platform through which users can access multiple services. From a system-analysis perspective, the important functions include user authentication, service selection, location selection, request submission, provider matching, real-time tracking, payment, rating and feedback, transaction history, and customer support.

These functions create an interconnected service lifecycle. A user does not simply perform one independent action; rather, a sequence of activities takes place from the initial application launch to the completion of a service. Consequently, a failure at one stage may influence later stages.

For example, if the application does not respond properly while a user is selecting a service, the user may be unable to submit a request. If location information is inaccurate, the system may process an incorrect pickup point. If a provider cancels after accepting a request, the user may need to repeat the booking process. If the final fare is not sufficiently clear, the completed transaction may result in a complaint. These relationships are important for understanding the five identified problems.

3.2.1.2 Review of Service Procedures

The existing Pathao service workflow can broadly be represented as follows:

1. The user opens the Pathao application.
2. The user logs in or registers an account.
3. The user selects a required service.
4. The user provides the required service information.
5. The system validates and processes the request.
6. Location and availability information are considered.
7. A suitable service provider is identified.
8. The provider accepts the request.
9. The service is performed and tracked.
10. Payment is processed.
11. The transaction is recorded.
12. The user can provide feedback or rating.
13. Customer support can be contacted if a problem occurs.

This procedure demonstrates why reliability is important throughout the complete service lifecycle. The system must not only perform individual functions correctly but must also maintain consistency as information moves from one process to another.

3.2.1.3 Review of Digital Forms and User Inputs

Unlike a traditional paper-based system, most information in Pathao is collected through digital interfaces. Important user inputs include:

- Mobile number and authentication information
- Pickup location
- Destination
- Service type
- Restaurant or merchant information where applicable
- Delivery address
- Payment information
- Ratings and feedback
- Complaint information

The quality of these inputs directly affects subsequent processing. For example, an incorrectly selected pickup location can affect provider matching, while incomplete complaint information can make issue resolution more difficult.

Therefore, the design of user-input interfaces is considered an important part of the system analysis.

3.2.1.4 Technology and Industry Considerations

The analysis also considers technologies commonly associated with modern on-demand service platforms, including GPS-based positioning, real-time tracking, cloud-based application services, database management, notification systems, digital payment processing, analytics, and automated customer-support tools.

These technologies provide possible directions for improving the five identified problems. However, the purpose of this project is not to assume that Pathao currently lacks a particular technology. Instead, the technologies are considered as possible mechanisms for improving reliability, transparency, and efficiency within the existing system.

3.2.2 Application-Based Observation

Observation provides information that may not be obtained through documentation alone. In a traditional organizational system study, an analyst may visit the organization's workplace and observe employees performing their responsibilities. For this project, direct access to Pathao's internal operational facilities is not assumed. Therefore, the observation component focuses on the publicly accessible user-facing application workflow.

The purpose of the observation is to understand how the system behaves from the perspective of a normal user and how the different service stages are connected.

3.2.2.1 User Authentication

The first stage of the interaction is user authentication. The application must identify the user before providing access to account-related services. This stage establishes the user session through which later activities are associated with the appropriate account.

3.2.2.2 Service Selection

After accessing the application, the user selects the required service. Because several services are available within the same application ecosystem, the system must clearly distinguish between the different service workflows.

An organized service-selection interface is therefore important for usability and operational accuracy.

3.2.2.3 Location and Service Information

For location-dependent services, users must provide pickup and destination information. This stage relies heavily on location services and map interfaces.

The observation of this process is particularly relevant to **Problem 02: Location Tracking and Map Accuracy Issues**, because the accuracy of the selected location can affect provider assignment, route information, and the user's ability to coordinate with the service provider.

3.2.2.4 Request Processing and Provider Matching

After the user submits the request, the system processes the information and attempts to identify an appropriate provider.

This stage is directly related to **Problem 03: Booking Reliability and Cancellation Problems**. A reliable system should provide clear status information and should respond appropriately when a provider is unavailable or cancels the request.

3.2.2.5 Tracking and Notifications

Once a service provider is assigned, the user receives status information and, where applicable, location updates. Tracking is therefore an important part of the overall service experience.

Any inconsistency in application performance or location information can affect the quality of this stage.

3.2.2.6 Payment and Transaction Completion

After service completion, payment is processed and the transaction is recorded. The user can then provide feedback or rating.

This stage is particularly relevant to **Problem 04: Fare Transparency and Pricing Confusion**, because users need to understand the amount charged and the relationship between the displayed estimate and the final transaction.

3.2.2.7 Customer Support

The final supporting component examined is customer support. If users experience booking problems, payment disputes, location issues, service dissatisfaction, or other difficulties, they require an effective mechanism for reporting and resolving those issues.

This observation is directly associated with **Problem 05: Customer Support and Complaint Resolution Delays**.

3.2.3 Stakeholder Interviews

Interviews are useful because they allow system analysts to obtain qualitative information that may not be fully represented by application observation. Different stakeholders may experience the same system differently. A customer may be concerned primarily with convenience and reliability, while a driver may be more concerned with location accuracy, request allocation, and communication.

For this project, the following interview questions are designed as a structured research instrument. They are not presented as interviews with Pathao employees or management unless such interviews are actually conducted.

3.2.3.1 Interview Questions for Pathao Users

Question 1: How frequently do you use Pathao services?

Purpose:

To identify the respondent's level of interaction with the platform.

Question 2: Have you experienced slow loading, lag, or unexpected application behavior?

Purpose:

To investigate the frequency and perceived severity of application-performance problems.

Question 3: Have you experienced incorrect pickup locations or inaccurate driver tracking?

Purpose:

To investigate GPS and map-related difficulties.

Question 4: Have you experienced a cancellation after requesting or booking a ride?

Purpose:

To examine booking reliability and cancellation-related problems.

Question 5: Do you clearly understand the fare shown before and after completing a service?

Purpose:

To assess users' perception of pricing transparency.

Question 6: Have you ever contacted Pathao support regarding a service problem?

Purpose:

To determine how frequently users depend on customer support.

Question 7: If you contacted customer support, how satisfied were you with the resolution process?

Purpose:

To evaluate the effectiveness and responsiveness of complaint handling.

Question 8: Which of the identified problems affects your overall experience the most?

Purpose:

To establish a relative priority among the five problems.

Question 9: What single improvement would make Pathao more reliable for you?

Purpose:

To identify user expectations that may contribute to proposed-system requirements.

3.2.3.2 Interview Questions for Drivers or Delivery Partners

The service-provider perspective is also important because the Pathao ecosystem depends on coordination between users and drivers or delivery partners.

1. How reliable is the request-notification process?
2. Do you frequently experience inaccurate customer pickup locations?
3. How often do customers cancel after you accept a request?
4. Do location or navigation problems affect your ability to complete trips efficiently?
5. Are service and fare-related information sufficiently clear?
6. What technical problem affects your work most frequently?
7. What improvement would make the application easier for you to use?

These questions can help determine whether a problem originates primarily from the customer side, provider side, system side, or the interaction between these components.

3.2.4 User Questionnaires

A questionnaire provides a structured method for collecting information from a larger number of respondents. The reference report uses several questionnaire formats, including dichotomous questions, ranking scales, rating scales, fill-in-the-blank questions, and multiple-choice questions.

For the Pathao project, the questionnaire is specifically designed around the five identified problems.

A. Dichotomous Questions

1. Have you experienced slow loading or lag while using Pathao?

- Yes
- No

2. Have you experienced an incorrect pickup location or inaccurate GPS tracking?

- Yes
- No

3. Have you experienced a cancellation after requesting a ride?

- Yes
- No

4. Have you ever been confused about the fare or additional charges?

- Yes
- No

5. Have you experienced difficulty getting a satisfactory response from customer support?

- Yes
- No

B. Rating-Scale Questions

Scale: 1 = Very Dissatisfied, 5 = Very Satisfied

1. How would you rate the overall performance of the Pathao application?

1 — 2 — 3 — 4 — 5

2. How satisfied are you with GPS and live-location accuracy?

1 — 2 — 3 — 4 — 5

3. How satisfied are you with the reliability of the booking process?

1 — 2 — 3 — 4 — 5

4. How clearly does Pathao communicate “fare” information?

1 — 2 — 3 — 4 — 5

5. How satisfied are you with customer support?

1 — 2 — 3 — 4 — 5

C. Multiple-Choice Questions

1. How frequently do you use Pathao?

- (a) Daily
- (b) Several times a week
- (c) Several times a month
- (d) Occasionally

2. Which Pathao service do you use most frequently?

- (a) Ride
- (b) Food
- (c) Parcel
- (d) Courier

3. Which problem affects your experience the most?

- (a) Application performance
- (b) GPS and tracking
- (c) Booking and cancellation
- (d) Fare transparency
- (e) Customer support

4. Which improvement should receive the highest priority?

- (a) Faster and more stable application
- (b) More accurate GPS and tracking
- (c) More reliable booking and reassignment

- (d) Clearer fare calculation
- (e) Faster complaint resolution

5. How would you describe your overall experience with Pathao?

- (a) Very poor
- (b) Poor
- (c) Average
- (d) Good
- (e) Excellent

3.3 Information Representation and Analysis

After information is gathered, it must be organized into a form that allows meaningful interpretation. Raw observations and questionnaire responses alone may not clearly demonstrate the relative importance of different problems. Therefore, graphical representation and system modeling are used to identify patterns and relationships.

The reference report converts questionnaire responses into charts and then develops DFDs to represent the existing system. A similar analytical approach is adopted for Pathao, but the representations are designed specifically around the five identified problems.

3.3.1 Proposed Questionnaire Charts

Once actual questionnaire responses are collected, the following charts should be generated.

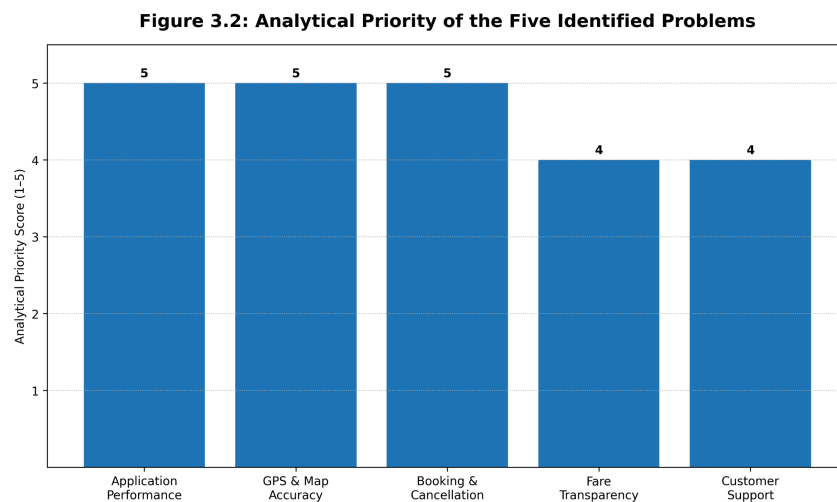


Figure 3.2: User Experience with Application Performance

This chart will show the percentage of respondents who have experienced application lag, slow loading, or other performance-related difficulties.

Figure 3.3: User Experience with GPS and Tracking

This chart will represent the proportion of respondents who have experienced inaccurate pickup locations, incorrect positioning, or tracking difficulties.

Figure 3.4: Booking Cancellation Experience

This chart will represent the percentage of respondents who have experienced cancellations after making a booking.

Figure 3.5: Perception of Fare Transparency

This chart will illustrate how clearly users understand the estimated and final fare.

Figure 3.6: Customer Support Satisfaction

This chart will represent users' satisfaction with support accessibility, response time, and problem resolution.

Figure 3.7: Most Significant Problem Identified by Users

This comparative chart will rank the five problems according to the respondents' selected priorities.

Note: No numerical percentages are assigned at this stage because actual questionnaire responses have not yet been collected. The final report should replace this placeholder stage with the actual results obtained from respondents.

3.3.2 Problem Impact Analysis

In addition to questionnaire charts, the five problems can be compared according to their likely effect on the overall service lifecycle.

Table 3.1: Comparative Impact of Identified Problems

Problem	User Impact	Operational Impact	System Impact	Priority
Inconsistent Application Performance	High	High	High	Very High
GPS and Map Accuracy Issues	High	High	High	Very High
Booking and Cancellation Problems	Very High	High	High	Very High
Fare Transparency Issues	High	Medium	Medium	High
Customer Support Delays	High	High	Medium	High

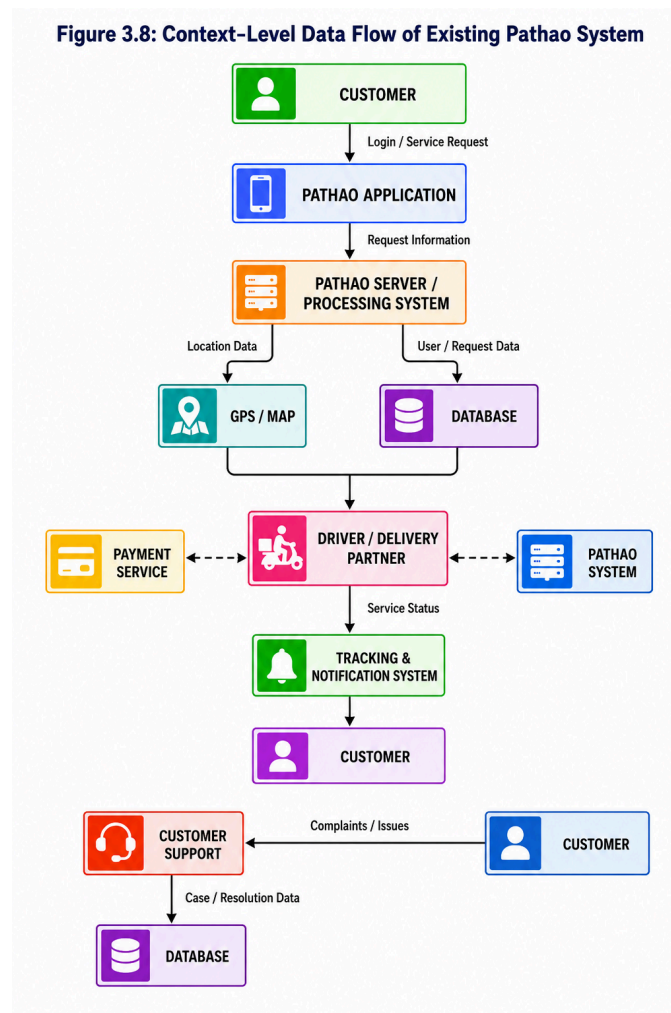
The table indicates that booking reliability and location accuracy have particularly broad effects because they influence the core service-delivery process. Application performance is also critical because it affects the user's interaction with multiple modules simultaneously.

Fare transparency and customer support remain important because they influence customer trust and post-service satisfaction.

3.3.3 Data Flow Diagram of the Existing Pathao System

A Data Flow Diagram (DFD) represents the movement of information through a system and identifies the processes that transform that information. The reference report describes DFDs as a means of representing data flows and corresponding transformations within a system.

For Pathao, the existing system can be represented through the interaction of the customer, application, server/database environment, location services, service provider, payment system, and customer support.



Description of Existing-System Data Flow

The process begins when a customer interacts with the Pathao application and submits a service request. The application sends the relevant information to the server-side processing environment.

The system uses location information and stored user or service-provider data to process the request. GPS and map services provide location-related information, while the database maintains information required for account management, requests, transactions, and service records.

After processing the request, an appropriate driver or delivery partner is identified. Service status and tracking information are then communicated to the customer through the application and notification mechanisms.

Once the service is completed, payment information is processed and the transaction is recorded. The customer may subsequently submit feedback or a rating.

If a problem occurs during any stage, the customer-support component becomes involved. Relevant information regarding the transaction or complaint can be used to investigate the issue and provide a resolution.

This model demonstrates that the five identified problems are distributed across several parts of the system rather than being limited to a single module.

3.3.4 Problem-to-Process Mapping

Table 3.2: Relationship Between Problems and System Processes

Problem	Main Affected Process	Secondary Processes
Application Performance	Application Access / Request Processing	Tracking, Payment, Support

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GPS and Map Accuracy	Location Processing	Matching, Navigation, Tracking
Booking Reliability	Provider Matching	Notifications, Service Execution
Fare Transparency	Fare Calculation / Payment	Support, Transaction History
Customer Support Delays	Complaint Management	Database, Transaction Records

This mapping demonstrates why improvements should not be implemented as isolated features. Changes to one process may require corresponding changes to other connected processes.

3.4 Requirements Analysis

Requirements analysis converts the problems identified during system analysis into specific expectations for the proposed system. A requirement describes what the improved system should accomplish or what quality standard it should maintain.

The requirements for the Pathao improvement project are derived directly from the five identified problems.

3.4.1 Functional Requirements

The proposed system should provide the following functional capabilities:

FR-01: Performance Monitoring

The system should continuously monitor application and service response performance and identify abnormal delays.

FR-02: Location Validation

The system should validate user-selected pickup locations and provide clearer location confirmation before a request is submitted.

FR-03: Improved Provider Matching

The system should select suitable service providers based on location, availability, and relevant service conditions.

FR-04: Cancellation Handling

The system should detect cancellations and, where possible, initiate a new provider-matching process without requiring the user to restart the entire booking procedure.

FR-05: Fare Breakdown

The system should provide a clear breakdown of relevant fare components and present the final transaction amount transparently.

FR-06: Complaint Registration

The system should allow users to submit complaints with relevant trip, order, or transaction information.

FR-07: Complaint Tracking

The system should assign a status to each complaint so users can monitor its progress.

FR-08: Automated Issue Classification

The system should categorize common complaints to help customer-support personnel process them more efficiently.

FR-09: Notification Management

The system should provide timely notifications regarding booking status, provider assignment, cancellation, payment, and complaint updates.

FR-10: Performance and Service Reports

The system should generate reports that allow administrators to identify recurring problems and monitor system performance.

3.4.2 Non-Functional Requirements

Performance

The proposed system should provide responsive interaction and should be capable of maintaining acceptable performance during periods of high demand.

Reliability

The system should minimize service interruptions and maintain consistent processing of requests.

Accuracy

Location, fare, booking, payment, and transaction information should be processed accurately.

Scalability

The system should be capable of supporting increased numbers of users and transactions without significant degradation in service quality.

Security

User, transaction, location, and account information should be protected against unauthorized access and inappropriate use.

Usability

The improved functions should remain understandable to ordinary users and should not unnecessarily complicate the existing workflow.

Maintainability

The system should be designed so that individual modules can be monitored, updated, and maintained without disrupting unrelated services.

3.4.3 User Requirements

From the perspective of an ordinary customer, the improved system should:

- Load quickly and respond consistently.
- Provide accurate pickup and destination information.
- Make booking status easy to understand.
- Reduce the inconvenience caused by cancellations.
- Explain fare information clearly.
- Provide accessible customer support.
- Allow users to track complaint progress.
- Maintain a consistent service experience.

From the perspective of drivers and delivery partners, the system should:

- Provide accurate customer locations.
- Deliver service requests reliably.
- Reduce unnecessary cancellation-related inefficiency.
- Provide clear service information.
- Improve communication with customers.

From the organizational perspective, the system should:

- Improve operational efficiency.
- Reduce repetitive complaints.
- Provide useful performance data.
- Support scalability.
- Improve customer satisfaction and retention.

3.5 Cost-Benefit Analysis

Cost-benefit analysis evaluates whether the expected advantages of improving the system justify the costs associated with development, implementation, operation, and maintenance. The reference report uses cost classification, annual investment estimation, and break-even analysis to evaluate the economic feasibility of the proposed system.

For Pathao, the proposed improvements involve software development, infrastructure optimization, testing, monitoring, location-service improvements, support-system enhancement, and ongoing maintenance.

Because confidential internal financial information from Pathao is not available for this academic project, the following monetary values are **illustrative estimates for academic analysis only**.

3.5.1 Types of Costs

Tangible Costs

Tangible costs are expenses that can be directly measured. For the proposed system, these may include:

- Software development.
- Server and cloud-resource upgrades.
- Database optimization.
- Monitoring tools.
- Testing infrastructure.
- Development and maintenance personnel.

Intangible Costs

Intangible costs are more difficult to measure financially. Examples include:

- Temporary inconvenience during deployment.
- User adaptation.
- Organizational adjustment.
- Possible disruption during migration or updates.
- Training requirements.

Direct Costs

Direct costs are specifically associated with implementing the proposed improvements, such as software development and infrastructure upgrades.

Indirect Costs

Indirect costs may include technical support, staff training, documentation, monitoring, and administrative activities.

Fixed Costs

Fixed costs are primarily associated with initial implementation, such as system development, infrastructure configuration, and integration.

Variable Costs

Variable costs may increase according to system usage. Examples include cloud-resource consumption, notification services, database capacity, and support workload.

3.5.2 Illustrative Cost Estimation

Table 3.3: Estimated Initial Development Cost

Cost Component	Estimated Amount (BDT)
Application performance optimization	10,00,000
GPS and location-service enhancement	8,00,000
Booking and cancellation improvement	7,00,000

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Fare-transparency module	4,00,000
Customer-support enhancement	6,00,000
Testing and quality assurance	4,00,000
Monitoring and analytics infrastructure	3,00,000
Estimated Initial Investment	42,00,000

These figures are not claimed to represent Pathao's actual internal expenditure. They are included to demonstrate the cost-benefit methodology required for this academic system-analysis project.

Assume that the improved system produces an estimated annual measurable benefit of **BDT 16,00,000**, through reduced operational inefficiencies, lower repetitive support workload, improved retention, and better service utilization.

Assume annual maintenance and infrastructure expenses of **BDT 4,00,000**.

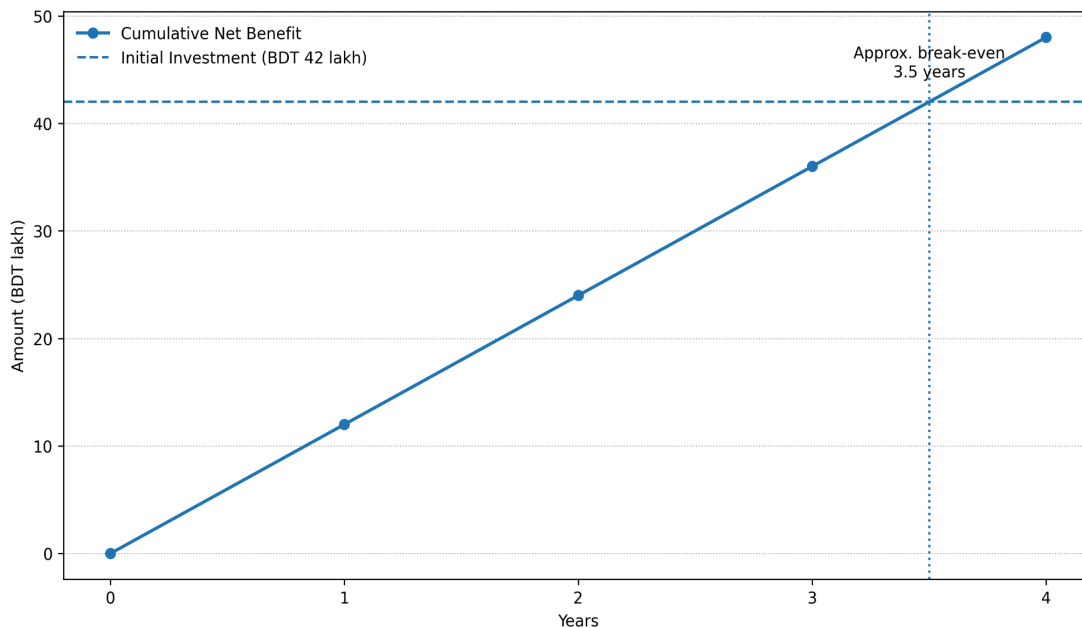
Therefore:

Net Annual Benefit = Annual Benefit – Annual Recurring Cost

= BDT 16,00,000 – BDT 4,00,000

= BDT 12,00,000

Figure 3.4: Illustrative Cost-Benefit and Break-Even Analysis



3.5.3 Break-Even Analysis

Break-even analysis identifies the point at which the cumulative financial benefits equal the initial investment.

The formula used is:

$$\text{Break-Even Period} = \text{Initial Investment} / \text{Net Annual Benefit}$$

Therefore:

$$\text{Break-Even Period} = 42,00,000 / 12,00,000$$

$$= 3.5 \text{ years}$$

Therefore, under the stated academic assumptions, the proposed improvement project would recover its initial investment in approximately **3 years and 6 months**.

The calculation demonstrates potential economic feasibility, but it should not be interpreted as a prediction of Pathao's actual financial return.

3.5.4 Expected Benefits

The proposed improvements are expected to produce both measurable and non-measurable benefits.

Operational Benefits

- Reduced system-related interruptions.
- More efficient provider matching.
- Improved location accuracy.
- Reduced repeated booking attempts.
- Better complaint-management efficiency.

Customer Benefits

- Faster application interaction.
- Greater confidence in location information.
- More reliable bookings.
- Better understanding of fare calculation.
- Improved complaint resolution.

Organizational Benefits

- Improved service quality.
- Better performance monitoring.
- More useful operational data.
- Improved customer retention.
- Stronger long-term scalability.

3.6 Detailed Feasibility Analysis

Detailed feasibility analysis examines the underlying causes of the identified problems and evaluates alternative approaches for improving the system. Rather than treating the five problems as independent defects, this section examines their relationship with system processes and determines which type of solution can address them most effectively.

The reference report follows a similar progression by performing root-cause analysis of its five major challenges before developing candidate systems and evaluating them.

3.6.1 Root Cause Analysis of Identified Problems

Problem 01: Inconsistent Application Performance

Root Cause

Application-performance problems can originate from several interacting factors, including inefficient application processes, heavy server requests, large amounts of real-time data, network limitations, insufficient resource optimization, or inefficient communication between application components.

Because Pathao combines several services within one application, the platform must handle different types of operations, including location processing, provider matching, notifications, payments, order-related activities, and account management. The complexity of these operations means that performance optimization is essential.

Performance Impact

Poor application performance can interrupt the normal transaction sequence. A delay while loading a service page or map can prevent users from completing their requests efficiently. If a request is interrupted during booking, the user may need to repeat the process.

Behavioral Impact

Repeated performance problems can reduce users' confidence in the application. Users who require an immediate service are particularly sensitive to delays because the value of an on-demand platform depends heavily on responsiveness.

Required Improvement

The system should incorporate stronger performance monitoring, optimized API communication, efficient data retrieval, caching where appropriate, scalable server resources, and automated identification of abnormal response times.

Problem 02: Location Tracking and Map Accuracy Issues

Root Cause

Location inaccuracies may occur because of limitations in GPS signals, weak network connectivity, inaccurate user-selected locations, delayed location synchronization, map-data limitations, or differences between the location detected by the device and the location selected by the user.

Performance Impact

An incorrect pickup location can result in difficulty finding the customer, inefficient routes, incorrect estimated arrival information, and unnecessary communication between the driver and customer.

The problem becomes particularly significant because location information influences provider matching and real-time tracking.

Behavioral Impact

Customers may become frustrated when the driver cannot identify the correct pickup point. Drivers may also experience delays and additional travel when the displayed location is inaccurate.

Required Improvement

The proposed system should introduce stronger location validation, clearer pickup-point confirmation, improved GPS synchronization, location accuracy indicators, and mechanisms for users to correct or confirm their position before submitting a request.

Problem 03: Booking Reliability and Cancellation Problems

Root Cause

Booking problems can be influenced by provider availability, matching efficiency, request allocation, demand conditions, provider acceptance behavior, and communication between the system and service provider.

When a provider accepts a request but later cancels, the customer may need to wait for another provider or repeat part of the booking process.

Performance Impact

Frequent cancellations increase waiting time and reduce the efficiency of the booking process. They can also increase system workload because the system must repeatedly process new provider-matching requests.

Behavioral Impact

Customers may perceive the service as unreliable when a confirmed or accepted booking does not proceed as expected. Repeated cancellation experiences can negatively influence customer loyalty.

Required Improvement

The proposed system should improve provider matching, monitor cancellation patterns, provide clearer booking status, and automatically attempt reassignment when appropriate.

Problem 04: Fare Transparency and Pricing Confusion

Root Cause

Fare confusion can occur when several pricing factors contribute to the final amount and the relationship between the estimated and final fare is not sufficiently clear to the user.

In an on-demand service environment, pricing may be affected by factors such as service type, distance, time, demand, or applicable charges. Therefore, simply displaying a final number may not always provide sufficient transparency.

Performance Impact

Unclear pricing can increase the number of fare-related complaints and create additional work for customer-support personnel.

Behavioral Impact

Users may feel uncertain or dissatisfied when they cannot understand why the final amount differs from their initial expectation. Lack of transparency can weaken trust in the platform.

Required Improvement

The proposed system should provide a clearer fare breakdown, identify applicable charges, explain important pricing factors, and provide a final transaction summary after service completion.

Problem 05: Customer Support and Complaint Resolution Delays

Root Cause

Complaint-resolution delays may result from high support demand, manual processing, insufficient case prioritization, repeated information requests, or difficulty connecting complaints with relevant trip and transaction information.

Performance Impact

An inefficient complaint system can create a backlog of unresolved cases. Support personnel may spend additional time locating transaction details that could otherwise be retrieved automatically.

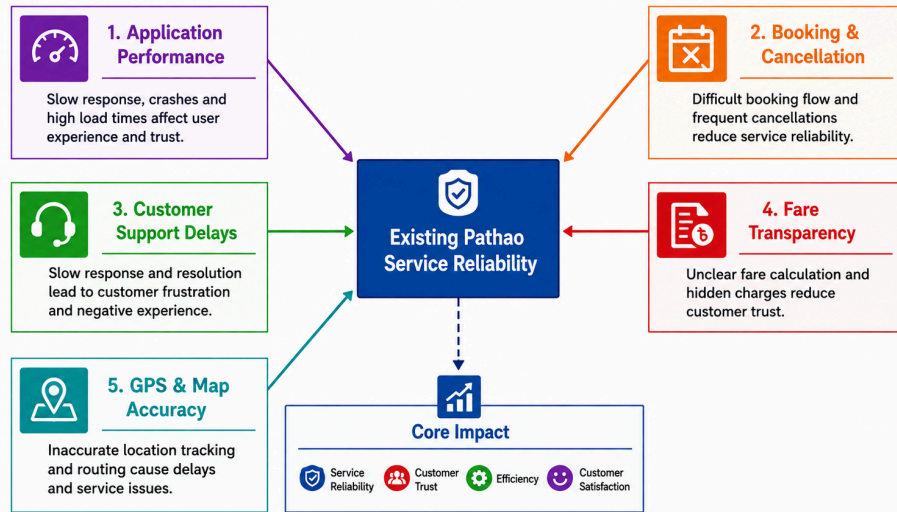
Behavioral Impact

Customers generally judge the quality of a service not only by whether problems occur but also by how effectively those problems are resolved. Delayed responses can therefore create dissatisfaction even after the original service has been completed.

Required Improvement

The proposed system should introduce centralized complaint management, automatic case classification, priority-based handling, direct access to relevant transaction information, status tracking, and automated updates for customers.

Figure 3.5: Root-Cause Relationship of the Five Identified Problems



3.6.2 Candidate Systems

After analyzing the five identified problems, two alternative improvement strategies are considered. The objective is to determine whether Pathao should improve its existing architecture incrementally or adopt a more intelligent and predictive system architecture.

Candidate System I: Integrated Incremental Improvement System

Candidate System I focuses on improving the existing Pathao platform without replacing its fundamental service structure. The objective is to strengthen the existing modules and introduce additional controls where problems have been identified.

Major Features

1. Performance Optimization Module

- Application performance monitoring.
- API optimization.
- Efficient data retrieval.
- Caching mechanisms where appropriate.
- Server-load monitoring.
- Automated performance alerts.

2. Location Accuracy Module

- GPS synchronization improvement.
- Pickup-location confirmation.
- Location accuracy indicator.
- Improved map interaction.
- Manual location correction.

3. Booking Reliability Module

- Improved provider matching.
- Booking-status visibility.
- Automatic reassignment after cancellation where feasible.
- Cancellation monitoring.

4. Transparent Fare Module

- Detailed fare breakdown.
- Clear presentation of applicable charges.
- Estimated versus final fare comparison.
- Digital transaction summary.

5. Customer Support Management Module

- Centralized complaint registration.
- Automatic complaint categorization.
- Priority assignment.
- Complaint-status tracking.
- Integration with trip and payment records.

Characteristics

Focus:

Improving the reliability and transparency of the existing Pathao service ecosystem through incremental modifications.

Technology Requirements:

- Existing application infrastructure with optimized components.
- Monitoring and analytics tools.

- Improved GPS integration.
- Matching-algorithm refinement.
- Customer-support management database.
- Notification services.

Major Advantage:

The system can be introduced progressively while preserving the application's familiar interface and major operational structure.

3.6.3 Candidate System II: Intelligent Predictive Service Management System

Candidate System II proposes a more advanced architecture in which analytics, predictive models, and automation are incorporated into the existing service-management environment.

The objective is not merely to respond to problems after they occur but to identify patterns and potentially predict problems before they significantly affect users.

Major Features

1. Predictive Performance Management

The system would monitor application response times, server demand, service traffic, and error patterns to identify potential performance degradation.

2. Intelligent Location Validation

The system could compare GPS signals, selected locations, map information, and other available location indicators to detect potentially inaccurate pickup points.

3. Predictive Provider Matching

Provider matching could consider not only physical proximity but also availability, expected arrival time, demand conditions, and historical response patterns.

4. Dynamic Fare Explanation

The system would provide a more detailed explanation of the factors contributing to the estimated fare and communicate relevant changes to the user.

5. Automated Customer-Support Assistant

An automated support component could classify common complaints, retrieve relevant transaction information, provide basic solutions, and transfer more complex cases to human support personnel.

Characteristics

Focus:

Creating a more intelligent, predictive, and data-driven service-management environment.

Technology Requirements:

- Advanced analytics.
- Predictive algorithms.
- Real-time monitoring.
- Machine-learning infrastructure where appropriate.
- Advanced location processing.
- Automated customer-support tools.
- Centralized analytics platform.

Major Advantage:

The system has greater long-term potential for automation, prediction, and large-scale optimization.

Major Limitation:

It requires greater development complexity, stronger data infrastructure, more extensive testing, and potentially higher implementation and maintenance costs.

3.6.4 Characteristics of Candidate Systems

Table 3.4: Comparative Characteristics of Candidate Systems

Criteria	Candidate System I	Candidate System II
Implementation Approach	Incremental improvement	Advanced transformation
Application Changes	Moderate	Significant
Initial Cost	Medium	High
Technical Complexity	Medium	High
User Adaptation	Low	Medium
Reliability Improvement	High	Very High
Scalability	High	Very High
Automation	Medium	Very High
Implementation Risk	Low	Medium–High

Long-Term Potential	High	Very High
Problem Coverage	High	Very High
Maintenance Complexity	Medium	High

3.6.5 Qualitative Evaluation Matrix

To identify the more appropriate candidate system, a qualitative evaluation is performed using criteria relevant to the Pathao project.

The selected criteria are:

1. User Friendliness
2. Reliability
3. Cost Effectiveness
4. Scalability
5. Implementation Risk
6. Problem Coverage
7. Maintainability
8. Future Expansion Capability

Each criterion is assigned a weight according to its importance to the project.

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Figure 3.6: Weighted-Evaluation Ratings of Candidate Systems

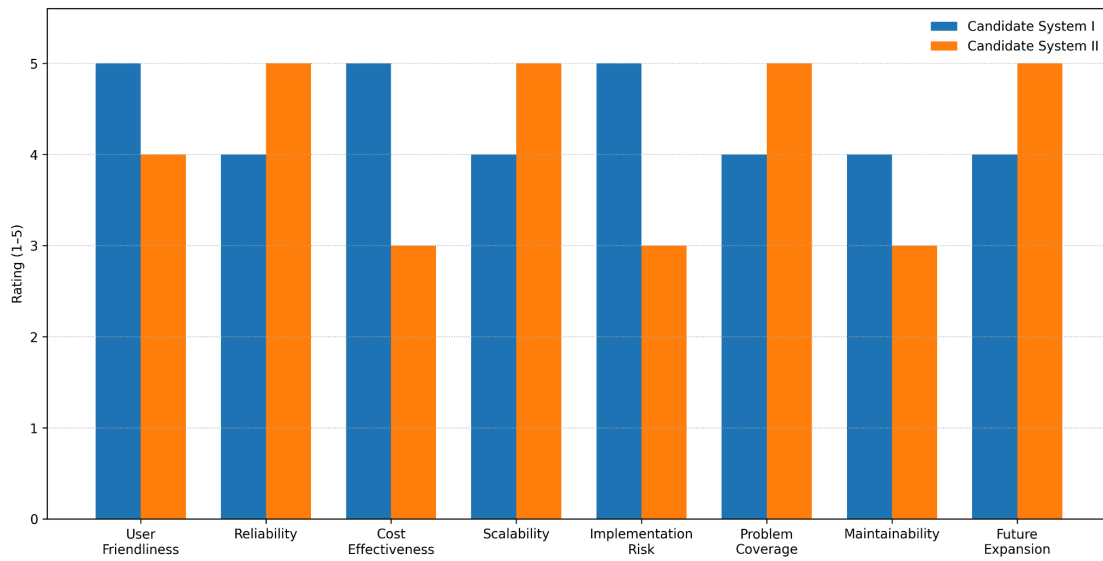


Table 3.5: Weighted Evaluation Criteria

Criterion	Weight
User Friendliness	10%
Reliability	20%
Cost Effectiveness	15%
Scalability	15%
Implementation Risk	10%
Problem Coverage	15%

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Maintainability	5%
Future Expansion Capability	10%
Total	100%

A score from 1 to 5 is assigned to each candidate:

- **1 = Very Poor**
- **2 = Poor**
- **3 = Average**
- **4 = Good**
- **5 = Excellent**

Table 3.6: Qualitative Evaluation Matrix

Criterion	Weight	Candidate I	Candidate II
User Friendliness	10	5	4
Reliability	20	4	5
Cost Effectiveness	15	5	3
Scalability	15	4	5
Implementation Risk	10	5	3

Problem Coverage	15	4	5
Maintainability	5	4	3
Future Expansion	10	4	5

3.6.6 Weighted Evaluation

The weighted score is calculated using:

$$\text{Weighted Score} = \sum (\text{Weight} \times \text{Rating})$$

where the rating ranges from 1 to 5 and the weight is expressed as a percentage.

Candidate System I

$$= (10 \times 5) + (20 \times 4) + (15 \times 5) + (15 \times 4) + (10 \times 5) + (15 \times 4) + (5 \times 4) + (10 \times 4)$$

$$= 50 + 80 + 75 + 60 + 50 + 60 + 20 + 40$$

$$= 435 / 500$$

Candidate System II

$$= (10 \times 4) + (20 \times 5) + (15 \times 3) + (15 \times 5) + (10 \times 3) + (15 \times 5) + (5 \times 3) + (10 \times 5)$$

$$= 40 + 100 + 45 + 75 + 30 + 75 + 15 + 50$$

$$= 430 / 500$$

Table 3.7: Final Weighted Evaluation

Candidate System	Score	Percentage	Position
Candidate System I	435 / 500	87%	1st
Candidate System II	430 / 500	86%	2nd

The difference between the two systems is relatively small. This is important because Candidate System II provides stronger long-term technological capabilities, while Candidate System I provides greater practicality and lower implementation risk.

3.6.7 Selection of the Best Candidate System

Based on the weighted evaluation, **Candidate System I: Integrated Incremental Improvement System** is selected as the preferred candidate for the current project.

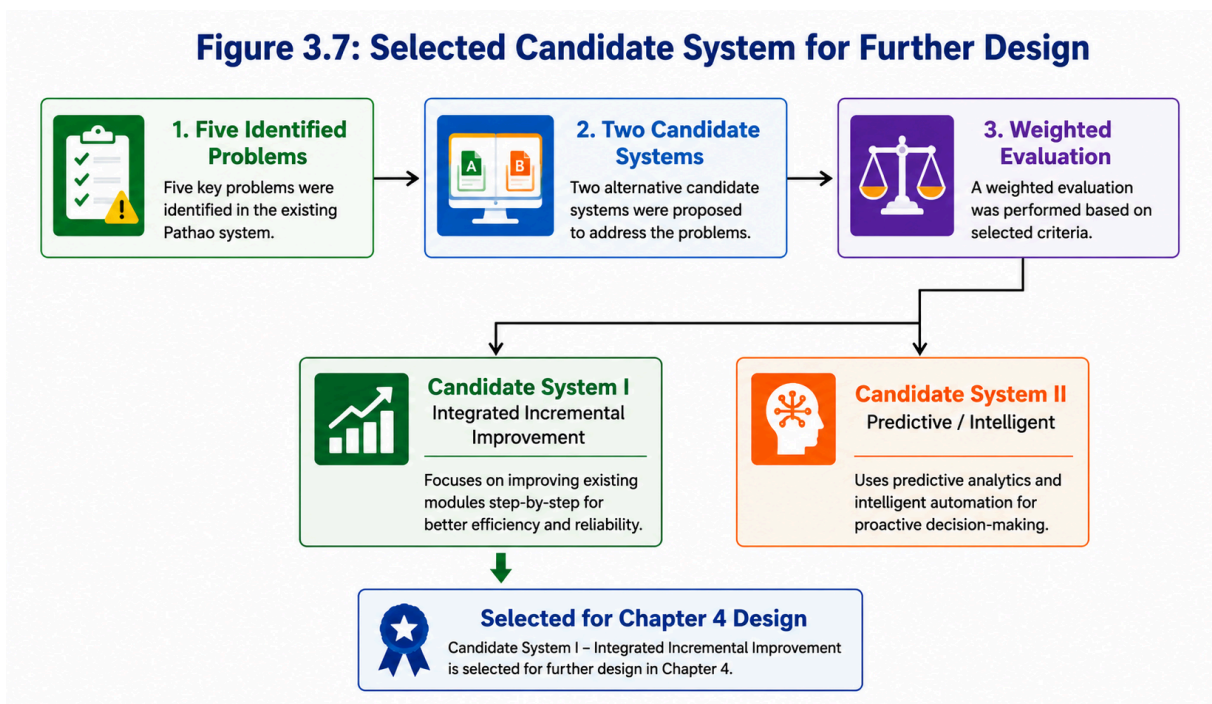
The primary reason for selecting Candidate System I is its balance between effectiveness and practicality. The objective of this project is to improve the identified weaknesses of the existing Pathao system rather than replace the entire platform. An incremental approach therefore reduces unnecessary disruption while still addressing the major problems identified during the analysis.

Candidate System I can directly address all five problems. Application monitoring can improve performance consistency, location validation can improve GPS reliability, improved matching and reassignment can reduce the impact of cancellations, transparent fare presentation can reduce pricing confusion, and centralized complaint management can improve customer-support efficiency.

Another advantage is user familiarity. Users would continue interacting with the same overall Pathao ecosystem rather than learning an entirely new platform. This reduces behavioral resistance and makes gradual deployment more practical.

Candidate System II remains a valuable long-term direction. Its predictive and intelligent capabilities could provide substantial benefits as the system matures. However, the greater complexity, infrastructure requirements, implementation risk, and maintenance requirements make it more appropriate as a future enhancement rather than the immediate solution selected for this project.

Therefore, Candidate System I provides the most balanced solution for the current stage of the project.



3.6.8 Recommended Improvement Strategy

The selected candidate should be implemented through a phased approach rather than introducing all changes simultaneously.

Phase 1: Performance and Monitoring

The first phase should establish system-performance monitoring and identify recurring response-time problems. Application and server performance should be measured continuously.

Phase 2: Location and Booking Reliability

The second phase should focus on GPS accuracy, pickup-point confirmation, provider matching, cancellation handling, and automatic reassignment.

Phase 3: Fare Transparency

The third phase should introduce a clearer presentation of fare components and transaction summaries.

Phase 4: Customer Support

The fourth phase should introduce centralized complaint management, automatic categorization, case tracking, and better integration with transaction records.

Phase 5: Continuous Monitoring

After deployment, the system should be continuously evaluated using performance metrics, complaint patterns, cancellation rates, location-related incidents, and user feedback.

This phased approach ensures that the proposed system is not treated as a one-time development project. Instead, it becomes a continuous improvement process in which system performance is monitored and further improvements are introduced based on measurable evidence.

3.7 Summary of Analysis Findings

The analysis performed in this chapter establishes a clear relationship between the five identified problems and the major processes of the Pathao system.

Table 3.8: Summary of Findings

Problem	Main Cause Area	Major Effect	Proposed Direction
Application Performance	Processing, server load, data handling	Slow or interrupted user interaction	Performance monitoring and optimization
GPS Accuracy	Location synchronization and input accuracy	Incorrect pickup/tracking	Location validation and synchronization
Booking Reliability	Provider availability and matching	Waiting and cancellations	Improved matching and reassignment
Fare Transparency	Pricing communication	Confusion and complaints	Fare breakdown and explanation
Customer Support	Case processing and information access	Delayed resolution	Centralized support management

The analysis demonstrates that the proposed system should not focus exclusively on one problem. The five areas should be treated as interconnected components of a single service ecosystem.

3.8 Conclusion

The analysis phase provided a comprehensive examination of the existing Pathao service environment and the five major problems identified in the Recognition of Need chapter. Through information gathering, application-based observation, stakeholder-oriented interviews, and questionnaire design, the study established a foundation for understanding the system from both technical and user perspectives.

The analysis also demonstrated how the five problems are connected to different processes within the Pathao service lifecycle. The existing-system DFD illustrated the flow of information among customers, the application, server and database components, location services, service providers, payment services, and customer support.

The cost-benefit analysis provided an approach for evaluating the economic feasibility of the proposed improvements, while the root-cause analysis identified performance management, location validation, provider matching, pricing communication, and complaint management as the key areas requiring improvement. Two candidate systems were then evaluated based on reliability, cost effectiveness, scalability, implementation risk, problem coverage, maintainability, and future expansion.

Based on the weighted evaluation, the **Integrated Incremental Improvement System** was selected as the preferred candidate because it provides a practical balance between system improvement and implementation risk while maintaining continuity with the existing Pathao platform.

The findings of this chapter provide the foundation for **Chapter 4: System Design**, where the selected system will be developed into detailed designs, including the proposed Data Flow Diagram, system architecture, database design, process models, and user-interface concepts. The design will remain aligned with the five problems identified in Chapter 1 to maintain consistency throughout the project.

Chapter 4

System Design

Introduction

System design is the stage in the system development process where the findings and requirements identified during the analysis phase are transformed into a structured model of the proposed system. It defines how the different components of the system will interact, how information will be processed and stored, and how users will interact with the system. A well-planned system design provides a clear technical foundation for implementing the proposed improvements while ensuring that the system remains consistent with the identified requirements.

In the previous chapters, the existing Pathao service environment was examined and five major problems were identified: **application performance management, location validation, service-provider matching, pricing communication, and complaint management**. The feasibility and detailed analysis conducted in Chapter 3 resulted in the selection of the **Integrated Incremental Improvement System** as the preferred candidate. The selected approach focuses on improving the existing Pathao platform rather than replacing it completely, thereby reducing implementation risk while addressing the major weaknesses identified during the analysis.

The proposed design therefore retains the core service structure of the existing Pathao platform while introducing improved processes, data management mechanisms, validation procedures, and communication features. The design covers the major activities involved in requesting and delivering a service, beginning with user authentication and service selection and continuing through location verification, provider assignment, fare communication, service completion, payment, rating, and complaint handling.

The purpose of this chapter is to translate the selected candidate system into a detailed system design. The proposed Data Flow Diagram illustrates the movement of information between users, the Pathao application, processing components, databases, location services, service providers, payment systems, and customer support. The database design defines the major entities and their relationships required to support the improved system. In addition, process

and form designs demonstrate how the proposed features can be incorporated into the user-facing application.

The design remains directly connected to the five problems identified in Chapter 1. Each proposed component is intended to address one or more of these problems while maintaining the usability and operational continuity of the existing Pathao platform.

Proposed System Overview

The proposed system is an enhanced version of the existing Pathao service platform. Instead of developing a completely separate application, the proposed design introduces incremental improvements to the existing service architecture. This approach allows the major existing functions of Pathao to remain available while improving the processes responsible for performance, location verification, provider assignment, pricing transparency, and complaint resolution.

The proposed system will support multiple Pathao services, including ride, food, parcel, and courier services. A centralized processing environment will receive user requests and coordinate the required information between the application, server, database, GPS and mapping services, service providers, payment services, and customer support.

A major improvement is the introduction of additional validation and decision-making stages before a service request is finalized. Location information will be validated before processing, available providers will be evaluated according to relevant factors such as proximity and availability, and estimated fares will be presented more clearly to users before confirmation. The system will also maintain structured complaint records so that customer issues can be tracked from submission to resolution.

The proposed system can therefore be summarized through the following major stages:

User Authentication → Service Selection → Information Entry → Location Validation → Request Processing → Provider Matching → Fare Calculation & Confirmation → Live Tracking → Service Completion → Payment → Rating/Feedback → Complaint Management & History

This workflow provides a more structured interaction between the major components of the Pathao service environment.

Objectives of the Proposed System

The proposed system design has been developed with the following objectives:

1. **Improve Application Performance:**
Introduce better request processing and system monitoring mechanisms to reduce unnecessary delays and improve application responsiveness during service requests.
2. **Improve Location Validation:**
Validate pickup and destination locations using GPS and map information before confirming a service request, reducing errors caused by inaccurate or ambiguous locations.
3. **Improve Service-Provider Matching:**
Develop a structured matching process that considers provider availability, proximity, service type, and other relevant factors when assigning a driver, rider, or courier.
4. **Improve Pricing Communication:**
Present fare estimates and relevant pricing information clearly before confirmation so that users have a better understanding of the expected cost.
5. **Improve Complaint Management:**
Introduce a structured complaint-management process that records complaints, assigns appropriate categories, tracks their status, and provides users with updates.
6. **Improve Data Management:**
Maintain centralized and organized records of users, service requests, providers, locations, payments, ratings, and complaints.
7. **Improve System Reliability:**
Establish better coordination among system components so that failures or inconsistencies in one process can be detected and managed more effectively.
8. **Maintain Scalability:**
Design the proposed system so that additional services, features, and users can be incorporated without requiring a complete restructuring of the system.

DFD of the Proposed Candidate System

The Data Flow Diagram (DFD) of the proposed candidate system represents how information will move through the improved Pathao platform. It demonstrates the interaction between external entities, processing activities, data stores, and supporting services.

Since **Candidate System I — Integrated Incremental Improvement System** was selected in Chapter 3, the proposed DFD focuses on incremental improvements to the existing Pathao environment. The major objective is to show how the five identified problems are addressed within the existing service workflow.

Context-Level DFD

At the highest level, the proposed Pathao system can be represented as a single central process interacting with the major external entities.

The primary external entities are:

- **Customer/User**
- **Driver/Rider/Courier**
- **Restaurant/Merchant**
- **Payment Service**
- **GPS/Map Service**
- **Customer Support Staff**

The customer provides authentication information, service requirements, pickup and destination information, payment information, ratings, feedback, and complaints. The system processes these inputs and returns service confirmations, fare information, provider information, tracking updates, payment status, and complaint status.

The service provider receives assignment information and sends availability, location, and service-status information to the system. GPS and map services provide location and route information, while payment services process and verify transactions. Customer support personnel access complaint information and update complaint-resolution status.

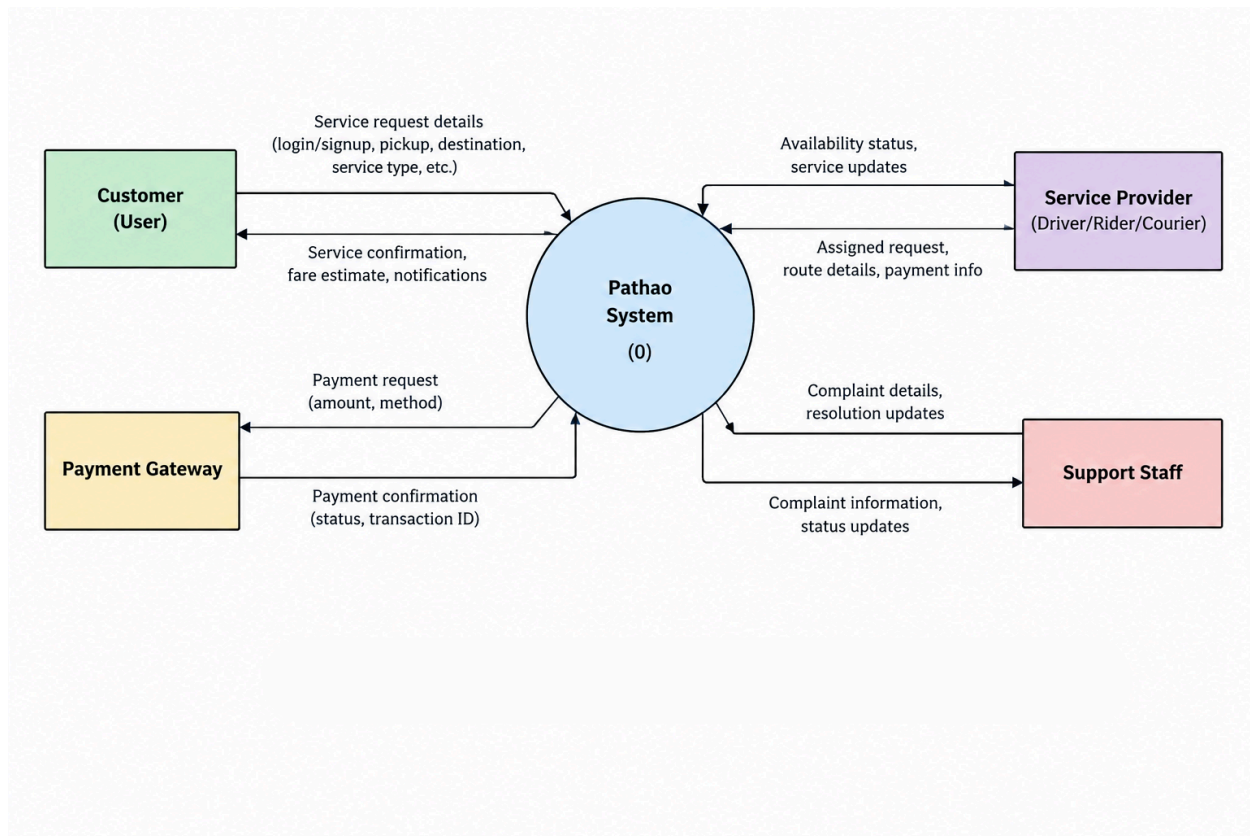


Figure: Context-Level DFD of the Proposed Pathao System

Level-1 DFD of the Proposed System

The Level-1 DFD divides the proposed Pathao system into several major processes. These processes represent the major stages through which a service request passes.

Process 1: User Authentication and Service Selection

The process verifies the user's account information and allows the user to select the required Pathao service. User information is retrieved from the User Database, while the selected service is passed to the request-processing stage.

Process 2: Request and Information Processing

The system receives information such as pickup location, destination, service type, restaurant information, parcel information, or other required details. The request is stored as a service request and forwarded for validation.

Process 3: Location Validation

The proposed system verifies the submitted location using GPS and mapping information. If the location is ambiguous or invalid, the system requests corrected information from the user. Once validated, the location information is stored and forwarded to the matching process.

This process directly addresses the **location validation problem** identified in Chapter 1.

Process 4: Service-Provider Matching

The system identifies suitable available providers based on factors such as service type, current availability, approximate distance, and location. The selected provider receives the service request and responds with acceptance or rejection.

This process addresses the **service-provider matching problem**.

Process 5: Fare Calculation and Communication

After validating the request, the system calculates or retrieves the estimated fare according to the selected service, distance, applicable pricing conditions, and other relevant factors. The estimated fare is presented to the customer before final confirmation.

This improves **pricing communication** by making the expected cost more visible during the booking process.

Process 6: Service Execution and Live Tracking

Once the provider accepts the request, the system activates the service and provides tracking information. The system continuously receives relevant location and status information and communicates appropriate updates to the customer.

Process 7: Payment Processing

After service completion, payment information is processed through the selected payment method. The system records the transaction and updates the service history.

Process 8: Rating and Feedback

After completion, the customer can provide a rating and feedback regarding the service. These records are stored for future evaluation and service-quality analysis.

Process 9: Complaint Management

If a customer experiences a problem, a complaint can be submitted through the application. The system records the complaint, categorizes it, assigns it to the appropriate support process, and tracks its status until resolution.

This process directly addresses the **complaint management problem**.

Process 10: Performance Monitoring and System History

The system stores relevant operational information, including service requests, provider performance, response times, location information, transactions, ratings, and complaints. These records can support performance monitoring and future system improvements.

This process contributes to addressing the **application performance management problem**.

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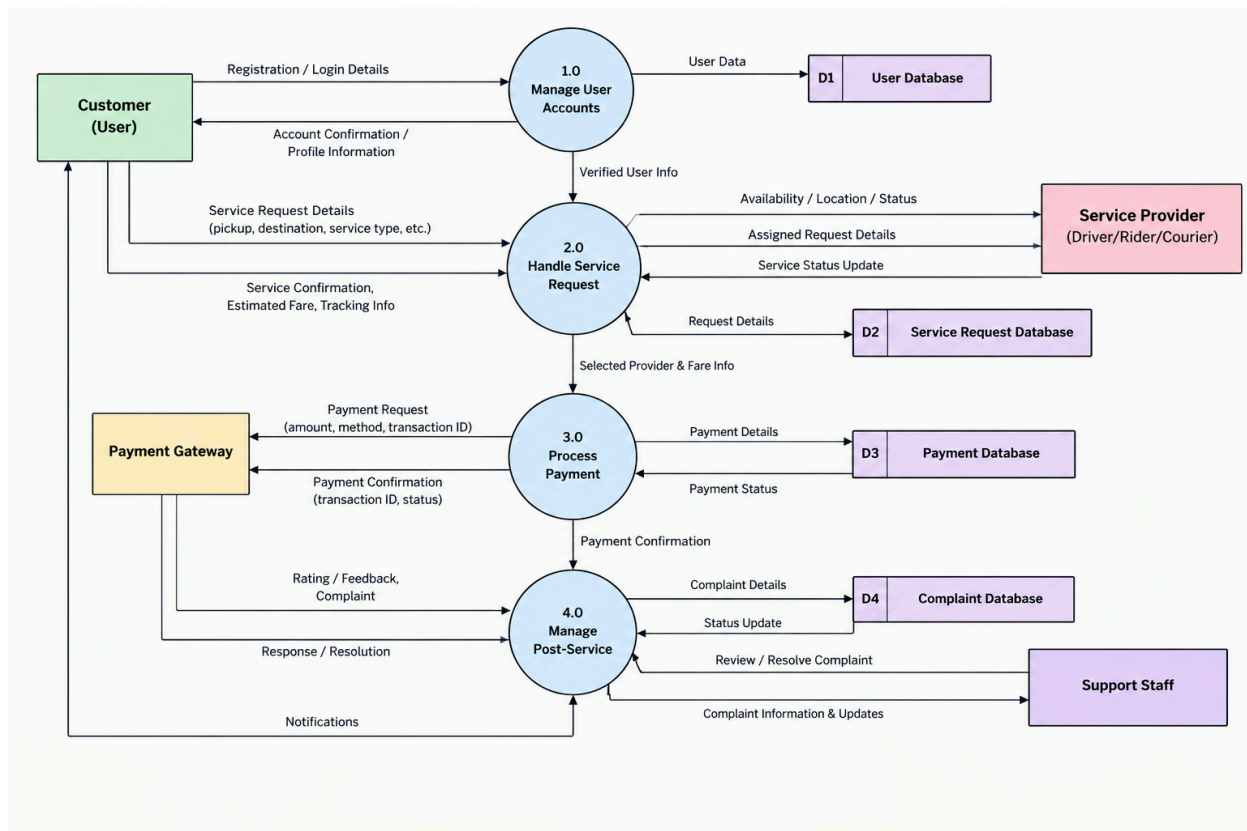


Figure: Level-1 DFD of the Proposed Pathao System

Proposed DFD Data Stores

The proposed system will require several major data stores:

User Information Database

Stores customer account information, contact details, authentication-related information, saved locations, and service history.

Service Request Database

Stores booking and service-request information, including request ID, service type, pickup location, destination, status, provider information, and timestamps.

Provider Information Database

Stores driver, rider, courier, and merchant-related information, including availability and relevant service records.

Location and Tracking Database

Maintains validated location information and relevant tracking records associated with active services.

Fare and Payment Database

Stores fare estimates, final fares, payment methods, transaction information, and payment status.

Feedback and Complaint Database

Stores ratings, feedback, complaints, complaint categories, assigned support personnel, resolution status, and relevant timestamps.

Proposed System Architecture

The proposed system follows a layered architecture to improve organization, maintainability, and scalability. The major layers are:

Presentation Layer

This layer represents the interfaces used by customers, service providers, and support personnel. It includes login, service selection, booking, fare display, tracking, payment, feedback, and complaint interfaces.

Application and Business Logic Layer

This layer processes service requests and contains the major system functions, including location validation, provider matching, fare calculation, request management, complaint processing, and performance monitoring.

Data Management Layer

This layer manages the databases required by the system. It stores user information, service requests, provider information, location data, payment records, ratings, feedback, and complaints.

External Service Integration Layer

The system communicates with external services such as GPS and mapping services and payment gateways. These integrations allow the proposed platform to obtain location information and process transactions efficiently.

Use Case Diagram

The Use Case Diagram represents the functional requirements of the proposed Pathao system from the perspective of its major users and external entities. It illustrates how different actors interact with the system and identifies the primary functions that each actor can perform. The diagram provides a high-level view of the proposed system's functionality and helps establish a clear relationship between user requirements and system operations.

The major actors in the proposed system are **Customer, Service Provider, and Support Staff**. The customer can register or log in, select a service, provide pickup and destination information, request a service, view estimated fare information, confirm a booking, track the service, make payment, provide ratings and feedback, and submit complaints. The Service Provider can log in, update availability, receive service requests, accept or reject requests, update service status, and complete assigned services. Support Staff can access complaints, review complaint details, update complaint status, communicate resolution information, and close resolved cases.

The proposed Use Case Diagram also reflects the five major improvement areas identified during the analysis phase. Location validation is incorporated into the service-request process, provider matching is associated with the booking and assignment process, fare communication is included before service confirmation, and complaint management is represented as a structured post-service function. Performance-related activities are supported through the system's request-processing and service-management functions.

The relationships among the use cases demonstrate how different functions are connected. For example, requesting a service requires the user to provide and validate location information before the system proceeds with provider matching and booking confirmation. Similarly, complaint submission can be followed by complaint tracking and resolution. These relationships

provide a functional representation of how the proposed system will operate from the users' perspective.

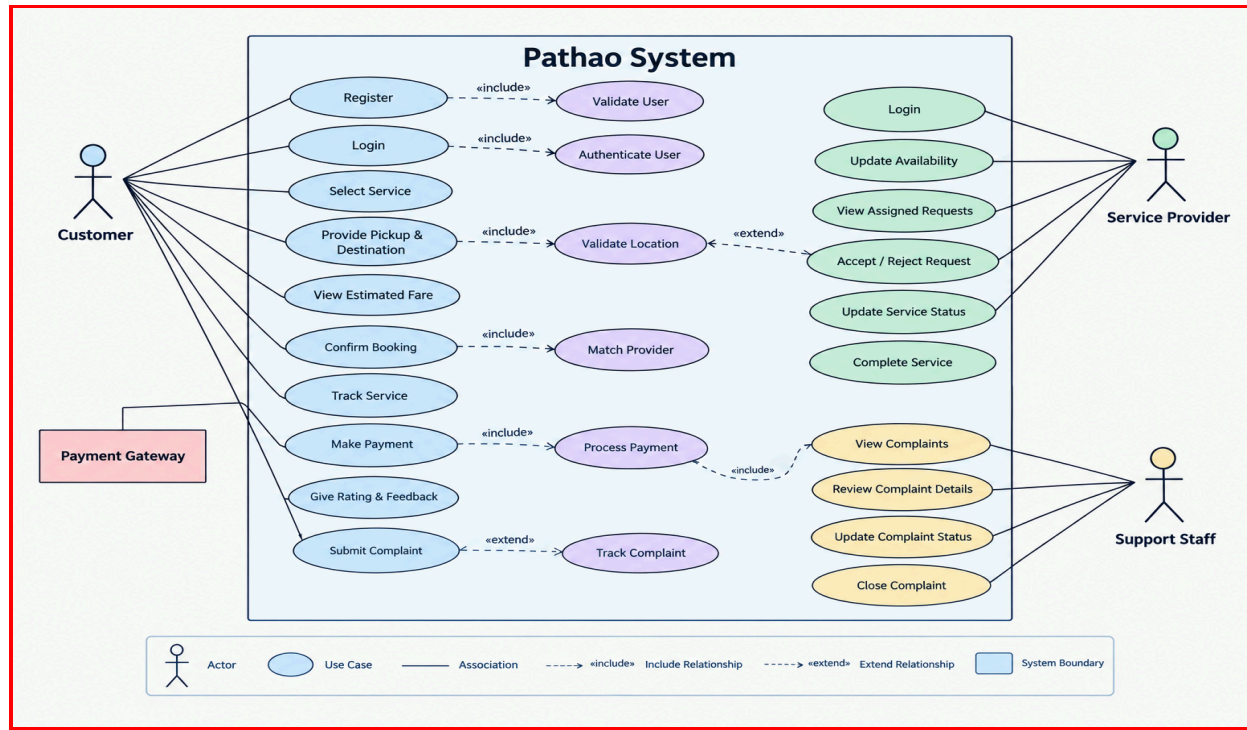


Figure : Use Case Diagram of the Proposed Pathao System

Proposed Process Models

The proposed process models describe how the major improved functions will operate.

Improved Service Request Process

The improved service-request process begins when the user selects a service and enters the required information. The system validates the information before creating the service request.

Process Flow:

User Login → Select Service → Enter Details → Validate Information → Validate Location → Calculate Fare → Display Fare → User Confirmation → Search Provider → Provider Acceptance → Service Activation

The validation stages reduce the likelihood of incorrect requests entering the provider-matching process.

Location Validation Process

The location validation process is designed to reduce errors caused by inaccurate pickup or destination information.

The system receives the location selected by the user and compares it with available GPS and map information. If the location is valid and sufficiently precise, it is confirmed. If the information is ambiguous, the user is requested to adjust or confirm the location before proceeding.

Process Flow:

Location Input → GPS/Map Verification → Accuracy Check → Valid Location? → Yes → Confirm Location

No → Request Location Correction → Revalidate

This process directly supports the improvement of the location validation problem.

Provider Matching Process

The provider matching process identifies an appropriate available service provider.

The system considers the selected service type, provider availability, approximate distance, current location, and relevant service information. A suitable provider is then selected and notified.

Process Flow:

Service Request → Identify Available Providers → Filter by Service Type → Compare Location & Availability → Select Suitable Provider → Send Request → Provider Accepts → Confirm Assignment

If the selected provider rejects the request, the system can continue searching for another suitable provider.

Fare Communication Process

The proposed fare process emphasizes transparency before service confirmation.

The system receives the service details and calculates an estimated fare based on the applicable factors. The estimate is displayed to the customer before confirmation.

Process Flow:

Service Details → Distance/Service Calculation → Fare Estimation → Display Estimated Fare → Customer Reviews → Confirm / Cancel → Continue Service

This reduces uncertainty and improves the customer's understanding of the expected service cost.

Complaint Management Process

The complaint-management process provides a structured method for submitting and resolving customer issues.

Process Flow:

Customer Complaint → Complaint Category Selection → Complaint Submission → Complaint ID Generation → Support Assignment → Investigation → Resolution → Customer Notification → Complaint Closure

The system maintains the complaint history so that unresolved cases can be monitored and reviewed.

Database Design

A well-structured database is essential for the proposed Pathao system because the platform needs to manage large volumes of user, service, location, provider, payment, feedback, and complaint information. The database design is intended to provide efficient data storage while maintaining data integrity, reducing unnecessary redundancy, and supporting future expansion.

Objectives of Database Design

The major objectives of the proposed database are:

1. **Efficient Data Storage:** Store all relevant system information in an organized structure.
2. **Data Integrity:** Maintain accurate and consistent records through appropriate validation and relationships.
3. **Reduced Redundancy:** Avoid unnecessary duplication of information.
4. **Fast Data Retrieval:** Allow system processes to access required information efficiently.
5. **Security:** Protect sensitive user and transaction-related information.
6. **Scalability:** Allow additional users, services, and features to be incorporated.
7. **Multi-User Support:** Support simultaneous operations by customers, providers, and administrative personnel.
8. **Reporting and Analysis:** Maintain structured records that can support performance monitoring and decision-making.

Entity Relationship Diagram

The proposed Entity Relationship (ER) diagram represents the major entities and relationships within the improved Pathao system. The primary entities include:

- User
- Service Request
- Provider
- Location
- Fare
- Payment
- Rating
- Complaint
- Support Staff
- Notification

A user can create multiple service requests, while each service request is associated with a particular service provider. Each request contains pickup and destination information and may have associated fare and payment records. After completion, a service request may generate a

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rating or complaint. Complaints are handled through the support-management component.

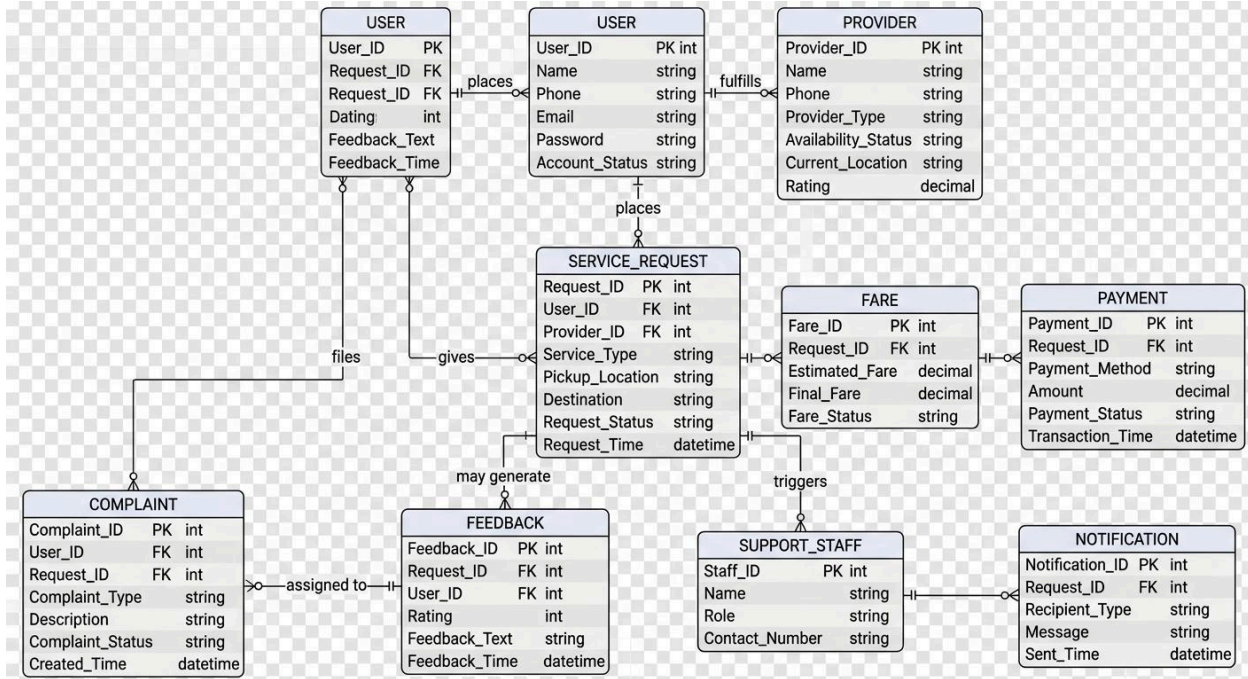


Figure: Entity Relationship Diagram of the Proposed Pathao System

Data Type and Size According to the ER Diagram

The following tables represent the major database structures proposed for the system.

User Table

Attribute	Type	Key
User_ID	INT	Primary Key
Name	VARCHAR(50)	—
Phone	VARCHAR(15)	—
Email	VARCHAR(100)	—
Password	VARCHAR(255)	—
Account_Status	VARCHAR(20)	—

Provider Table

Attribute	Type	Key
Provider_ID	INT	Primary Key
Name	VARCHAR(50)	—
Phone	VARCHAR(15)	—
Provider_Type	VARCHAR(30)	—
Availability_Status	VARCHAR(20)	—
Current_Location	VARCHAR(150)	—
Rating	DECIMAL(3,2)	—

Service Request Table

Attribute	Type	Key
Request_ID	INT	Primary Key
User_ID	INT	Foreign Key
Provider_ID	INT	Foreign Key
Service_Type	VARCHAR(30)	—
Pickup_Location	VARCHAR(150)	—
Destination	VARCHAR(150)	—
Request_Status	VARCHAR(30)	—
Request_Time	DATETIME	—

Fare Table

Attribute	Type	Key
Fare_ID	INT	Primary Key
Request_ID	INT	Foreign Key
Estimated_Fare	DECIMAL(10,2)	—
Final_Fare	DECIMAL(10,2)	—
Fare_Status	VARCHAR(20)	—

Payment Table

Attribute	Type	Key
Payment_ID	INT	Primary Key
Request_ID	INT	Foreign Key
Payment_Method	VARCHAR(30)	—
Amount	DECIMAL(10,2)	—
Payment_Status	VARCHAR(20)	—
Transaction_Time	DATETIME	—

Complaint Table

Attribute	Type	Key
Complaint_ID	INT	Primary Key
User_ID	INT	Foreign Key
Request_ID	INT	Foreign Key
Complaint_Type	VARCHAR(50)	—
Description	TEXT	—
Complaint_Status	VARCHAR(30)	—
Created_Time	DATETIME	—

Rating and Feedback Table

Attribute	Type	Key
Feedback_ID	INT	Primary Key
Request_ID	INT	Foreign Key
User_ID	INT	Foreign Key
Rating	INT	—
Feedback_Text	TEXT	—
Feedback_Time	DATETIME	—

Form and User Interface Design

Form design focuses on how users provide information to and receive information from the proposed system. Since the proposed solution is an incremental improvement of the existing Pathao platform, the design does not require a completely new interface. Instead, selected interfaces are improved to address the problems identified during analysis.

Login and Registration Form

The login and registration interface allows users to securely access the system. The proposed interface should provide clear input fields for phone number or email and password, together with account recovery and verification options.

Service Selection Interface

After authentication, users can select the required service, such as Ride, Food, Parcel, or Courier. The interface should provide clear service categories and reduce unnecessary navigation.

Location and Booking Form

The proposed booking interface will allow users to enter or select pickup and destination locations. GPS and map-based validation will provide additional confirmation before the booking proceeds.

The interface should display the selected locations clearly and provide an option to adjust an incorrect location.

Fare Display Interface

Before final confirmation, the system will display the estimated fare and relevant pricing information. This allows the customer to understand the expected cost before proceeding.

Provider and Live-Tracking Interface

After a provider accepts the request, the customer will be able to view provider information and the current service status. Live location information will be presented through the map interface.

Complaint Management Interface

The proposed complaint interface will allow users to select a complaint category, provide a description, submit supporting information where applicable, and track the status of the complaint.

Proposed System Features and Improvements

The proposed design introduces several improvements directly related to the five problems identified in Chapter 1.

Identified Problem	Proposed Design Improvement
Application Performance Management	Improved request processing, performance monitoring, and structured service workflow
Location Validation	GPS/map-based location verification before booking confirmation
Service-Provider Matching	Availability- and location-based provider matching
Pricing Communication	Clear estimated fare presentation before confirmation
Complaint Management	Categorized complaint submission, tracking, assignment, and resolution

These improvements are designed to work together rather than operate as isolated features. For example, accurate location validation improves provider matching, while improved provider matching contributes to faster service completion. Similarly, clear fare communication can reduce customer disputes, while structured complaint management provides a mechanism for resolving issues that still occur.

Security and Data Protection Considerations

Security is an important consideration in the proposed system because the platform handles personal information, location data, service records, and payment-related information.

The proposed design should therefore include appropriate authentication and authorization mechanisms to ensure that users can access only the information relevant to their accounts and roles. Sensitive information should be protected during storage and transmission. Database access should also be restricted according to user roles.

Location information should be handled carefully because it is directly associated with customer and provider activities. Access to location records should be limited to authorized processes and retained only as required for legitimate system functions.

Payment information should be processed through appropriate secure payment mechanisms rather than being unnecessarily stored within the application database. Complaint and support records should also be protected because they may contain personal or service-related information.

Regular system monitoring, error logging, backup procedures, and security updates should be incorporated into the system's operational design to improve overall reliability and protection.

Expected Benefits of the Proposed Design

The proposed system is expected to provide several benefits to both customers and service providers.

Improved User Experience

Clearer booking, location, fare, tracking, and complaint interfaces can make the overall service experience more understandable and convenient.

Improved Service Reliability

Better request processing and performance monitoring can reduce delays and improve system responsiveness.

More Accurate Location Processing

Location validation can reduce incorrect pickup or destination information and improve coordination between users and providers.

Better Provider Assignment

Improved matching can help connect service requests with suitable available providers more efficiently.

Greater Pricing Transparency

Displaying estimated fare information before confirmation can reduce confusion and improve customer confidence.

Improved Complaint Resolution

Structured complaint tracking can make it easier for support personnel to monitor and resolve customer issues.

Better Data Management

Centralized and structured data storage can improve record management and support future analysis.

Future Scalability

The modular nature of the proposed design allows additional services and improvements to be incorporated as the platform evolves.

Conclusion

The system design phase transforms the analytical findings from the previous chapters into a structured design for the proposed Pathao improvement system. The **Integrated Incremental Improvement System** selected in Chapter 3 has been used as the basis for designing the proposed processes, data flows, database structure, and user interfaces.

The proposed design directly addresses the five identified problems: application performance management, location validation, service-provider matching, pricing communication, and complaint management. The proposed DFD demonstrates how these improvements interact with the existing Pathao service environment, while the database design establishes a structured foundation for managing users, service requests, providers, locations, payments, feedback, and complaints.

The proposed process models and interface designs further demonstrate how the improved system can operate from the user's initial request through service completion and post-service support. Security and scalability considerations have also been incorporated to ensure that the design remains suitable for future development.

Overall, the proposed system provides a practical design framework for improving the existing Pathao platform without requiring a complete replacement of its core infrastructure. The designs developed in this chapter will serve as the foundation for the subsequent stages of implementation, testing, and evaluation.